
6-Port USB 3.2 Gen 2 Controller Hub

Highlights

- 6-Port USB 3.2 Gen 2 Smart Hub with:
 - Five Standard USB 3.2 Gen 2 downstream ports
 - One Standard USB 2.0 downstream port
 - Internal Hub Feature Controller enables:
 - USB to I²C/SPI/I²S/GPIO bridge endpoint support
 - USB to internal hub register write and read
- USB Link Power Management (LPM) support
- Programming of firmware image to external SPI memory device from USB host
- USB-IF Battery Charger revision 1.2 support on downstream ports (DCP, CDP, SDP)
- Enhanced OEM configuration options available through either OTP or external SPI memory
- Available in 100-pin (12 mm x 12 mm) VQFN RoHS compliant package
- Commercial (0°C to +70°C) and industrial (-40°C to +85°C) temperature range support

Target Applications

- Standalone USB Hubs
- Laptop Docks
- PC Motherboards
- PC Monitor Docks
- Multi-function USB 3.2 Gen 2 Peripherals

Key Benefits

- USB 3.2 Gen 2 compliant 10 Gbps, 5 Gbps, 480 Mbps, 12 Mbps, and 1.5Mbps operation
 - 5V tolerant USB 2.0 pins
 - 1.21V tolerant USB 3.2 Gen 2 pins
 - Integrated termination and pull-up/down resistors

- Supports battery charging of most popular battery powered devices on all ports
 - USB-IF Battery Charging rev. 1.2 support (DCP, CDP, SDP)
 - Apple® portable product charger emulation
 - Chinese YD/T 1591-2006/2009 charger emulation
 - European Union universal mobile charger support
 - Supports additional portable devices
- On-chip Microcontroller
 - Manages I/Os, VBUS, and other signals
- 96 kB RAM, 256 kB ROM
- 8 kB One-Time-Programmable (OTP) ROM
 - Includes on-chip charge pump
- Configuration programming via OTP Memory, SPI external memory or SMBus
- **FlexConnect**
 - The roles of the upstream and downstream ports are reversible on command
- **USB Bridging**
 - USB to I²C, SPI, I²S, and GPIO
- **PortSwap**
 - Configurable USB 2.0 differential pair signal swap
- **PHYBoost™**
 - Programmable USB transceiver drive strength for recovering signal integrity
- **VariSense™**
 - Programmable USB receive sensitivity
- **PortSplit**
 - USB 2.0 and USB 3.2 Gen 2 port operation can be split for custom applications using embedded USB 3.x devices in parallel with USB 2.0 devices
- Compatible with Microsoft Windows® 11, 10, 8, 7, XP, Apple OS X 10.4+, and Linux hub drivers
- Optimized for low-power operation and low thermal dissipation
- 100-pin VQFN package (12 mm x 12 mm)

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1.0 PREFACE

1.1 General Terms

TABLE 1-1: GENERAL TERMS

Term	Description
ADC	Analog-to-Digital Converter
Byte	8 bits
CDC	Communication Device Class
CSR	Control and Status Registers
DFP	Downstream Facing Port
DWORD	32 bits
EOP	End of Packet
EP	Endpoint
FIFO	First In First Out buffer
FS	Full-Speed
FSM	Finite State Machine
GPIO	General Purpose I/O
HS	Hi-Speed
HSOS	High Speed Over Sampling
Hub Feature Controller	The Hub Feature Controller, sometimes called a Hub Controller for short is the internal processor used to enable the unique features of the USB Controller Hub. This is not to be confused with the USB Hub Controller that is used to communicate the hub status back to the Host during a USB session.
I²C	Inter-Integrated Circuit
LS	Low-Speed
lsb	Least Significant Bit
LSB	Least Significant Byte
msb	Most Significant Bit
MSB	Most Significant Byte
N/A	Not Applicable
NC	No Connect
OTP	One Time Programmable
PCB	Printed Circuit Board
PCS	Physical Coding Sublayer
PHY	Physical Layer
PLL	Phase Lock Loop
RESERVED	Refers to a reserved bit field or address. Unless otherwise noted, reserved bits must always be zero for write operations. Unless otherwise noted, values are not guaranteed when reading reserved bits. Unless otherwise noted, do not read or write to reserved addresses.
SDK	Software Development Kit
SMBus	System Management Bus
UFP	Upstream Facing Port
UUID	Universally Unique Identifier
WORD	16 bits

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1.2 Buffer Types

TABLE 1-2: BUFFER TYPES

Buffer Type	Description
I	Input.
IS	Input with Schmitt trigger.
O12	Output buffer with 12 mA sink and 12 mA source.
OD12	Open-drain output with 12 mA sink
PU	50 μ A (typical) internal pull-up. Unless otherwise noted in the pin description, internal pull-ups are always enabled. Internal pull-up resistors prevent unconnected inputs from floating. Do not rely on internal resistors to drive signals external to the device. When connected to a load that must be pulled high, an external resistor must be added.
PD	50 μ A (typical) internal pull-down. Unless otherwise noted in the pin description, internal pull-downs are always enabled. Internal pull-down resistors prevent unconnected inputs from floating. Do not rely on internal resistors to drive signals external to the device. When connected to a load that must be pulled low, an external resistor must be added.
ICLK	Crystal oscillator input pin
OCLK	Crystal oscillator output pin
I/O-U	Analog input/output defined in USB specification.
I-R	RBIAS.
A	Analog.
AIO	Analog bidirectional.
P	Power pin.

1.3 Pin Reset States

The pin reset state definitions are detailed in [Table 1-3](#). Refer to [Section 3.1, Pin Assignments](#) for details on individual pin reset states.

TABLE 1-3: PIN RESET STATE LEGEND

Symbol	Description
AI	Analog input
AIO	Analog input/output
AO	Analog output
PD	Hardware enables pull-down
PU	Hardware enables pull-up
Y	Hardware enables function
Z	Hardware disables output driver (high impedance)
PU	Hardware enables internal pull-up
PD	Hardware enables internal pull-down

1.4 Reference Documents

1. *Universal Serial Bus Revision 3.2 Specification*, <http://www.usb.org>
2. *Battery Charging Specification*, Revision 1.2, Dec. 07, 2010, <http://www.usb.org>
3. *I²C-Bus Specification*, Version 1.1, http://www.nxp.com/documents/user_manual/UM10204.pdf
4. *I²S-Bus Specification*, <http://www.sparkfun.com/datasheets/BreakoutBoards/I2SBUS.pdf>
5. *System Management Bus Specification*, Version 1.0, <http://smbus.org/specs>

Note: Additional USB7206C resources can be found on the Microchip USB7206C product page at www.microchip.com/USB7206C.

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2.0 INTRODUCTION

2.1 General Description

The Microchip USB7206C hub is a low-power, OEM configurable, USB 3.2 Gen 2 hub controller with 6 downstream ports and advanced features for embedded USB applications. The USB7206C is fully compliant with the Universal Serial Bus Revision 3.2 Specification and USB 2.0 Link Power Management Addendum. The USB7206C supports 10 Gbps SuperSpeed+ (SS+), 5 Gbps SuperSpeed (SS), 480 Mbps Hi-Speed (HS), 12 Mbps Full-Speed (FS), and 1.5 Mbps Low-Speed (LS) USB downstream devices on five standard USB 3.2 Gen 2 downstream ports and only legacy speeds (HS/FS/LS) on one standard USB 2.0 downstream port.

The USB7206C supports the legacy USB speeds (HS/FS/LS) through a dedicated USB 2.0 hub controller that is the culmination of seven generations of Microchip hub feature controller design and experience with proven reliability, interoperability, and device compatibility. The SuperSpeed hub controller operates in parallel with the USB 2.0 controller, decoupling the 10/5 Gbps SS+/SS data transfers from bottlenecks due to the slower USB 2.0 traffic.

The USB7206C enables OEMs to configure their system using “Configuration Straps.” These straps simplify the configuration process assigning default values to USB 3.2 Gen 2 ports and GPIOs. OEMs can disable ports, enable battery charging and define GPIO functions as default assignments on power up removing the need for OTP or external SPI ROM.

The USB7206C supports downstream battery charging. The USB7206C integrated battery charger detection circuitry supports the USB-IF Battery Charging (BC1.2) detection method and most Apple devices. The USB7206C provides the battery charging handshake and supports the following USB-IF BC1.2 charging profiles:

- DCP: Dedicated Charging Port (Power brick with no data)
- CDP: Charging Downstream Port (1.5A with data)
- SDP: Standard Downstream Port (0.5A[USB 2.0]/0.9A[USB 3.2] with data)

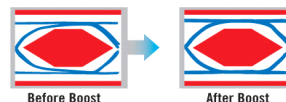
Additionally, the USB7206C includes many powerful and unique features such as:

The Hub Feature Controller, an internal USB device dedicated for use as a USB to I²C/SPI/GPIO interface that allows external circuits or devices to be monitored, controlled, or configured via the USB interface.

FlexConnect, which provides flexible connectivity options. One of the USB7206C’s downstream ports can be reconfigured to become the upstream port, allowing master capable devices to control other devices on the hub.

PortSwap, which adds per-port programmability to USB differential-pair pin locations. PortSwap allows direct alignment of USB signals (D+/D-) to connectors to avoid uneven trace length or crossing of the USB differential signals on the PCB.

PHYBoost, which provides programmable levels of Hi-Speed USB signal drive strength in the downstream port transceivers. PHYBoost attempts to restore USB signal integrity in a compromised system environment. The graphic on the right shows an example of Hi-Speed USB eye diagrams before and after PHYBoost signal integrity restoration. in a compromised system environment.



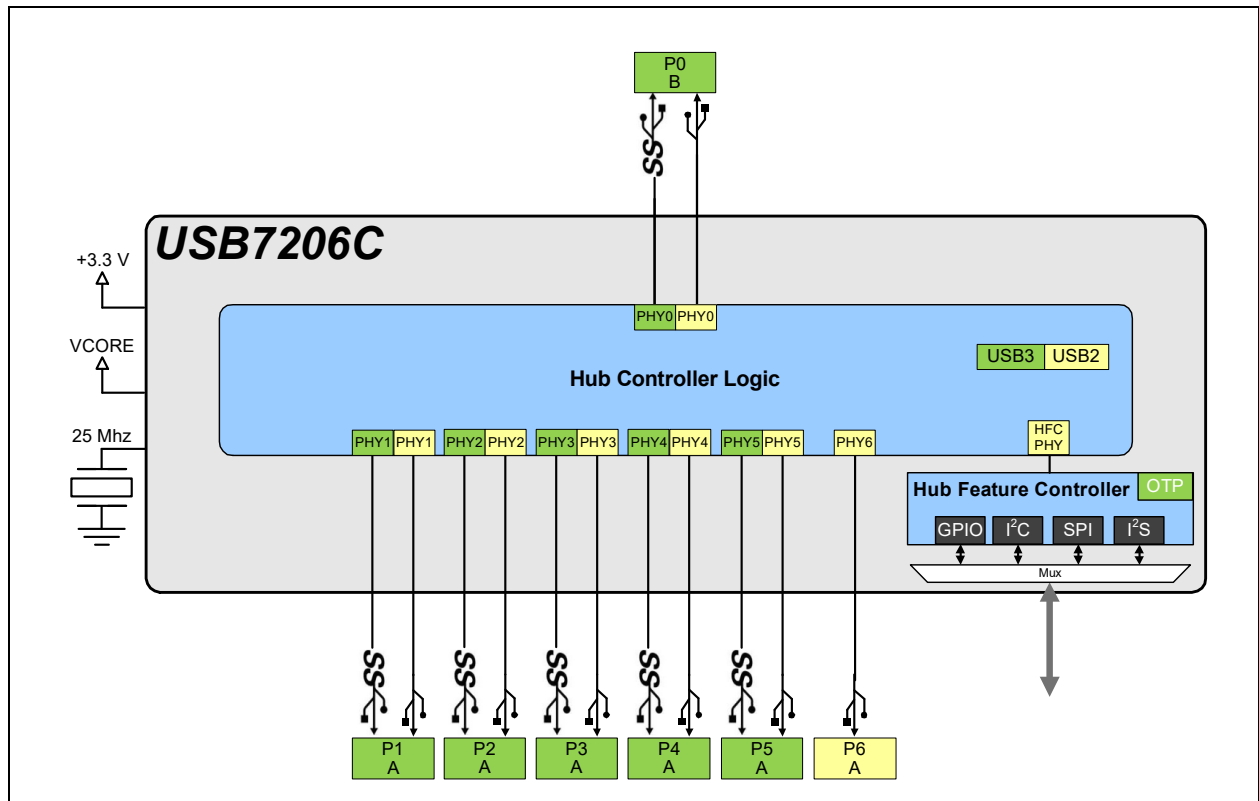
VariSense, which controls the Hi-Speed USB receiver sensitivity enabling programmable levels of USB signal receive sensitivity. This capability allows operation in a sub-optimal system environment, such as when a captive USB cable is used.

Port Split, which allows for the USB 3.2 Gen 2 and USB 2.0 portions of downstream ports 3, 4, and 5 to operate independently and enumerate two separate devices in parallel in special applications.

The USB7206C can be configured for operation through internal default settings. Custom OEM configurations are supported through external SPI ROM or OTP ROM. All port control signal pins are under firmware control in order to allow for maximum operational flexibility and are available as GPIOs for customer specific use.

The USB7206C is available in commercial (0°C to +70°C) and industrial (-40°C to +85°C) temperature ranges. An internal block diagram of the USB7206C in an upstream Type-B application is shown in [Figure 2-1](#).

FIGURE 2-1: USB7206C INTERNAL BLOCK DIAGRAM - UPSTREAM TYPE-B APPLICATION



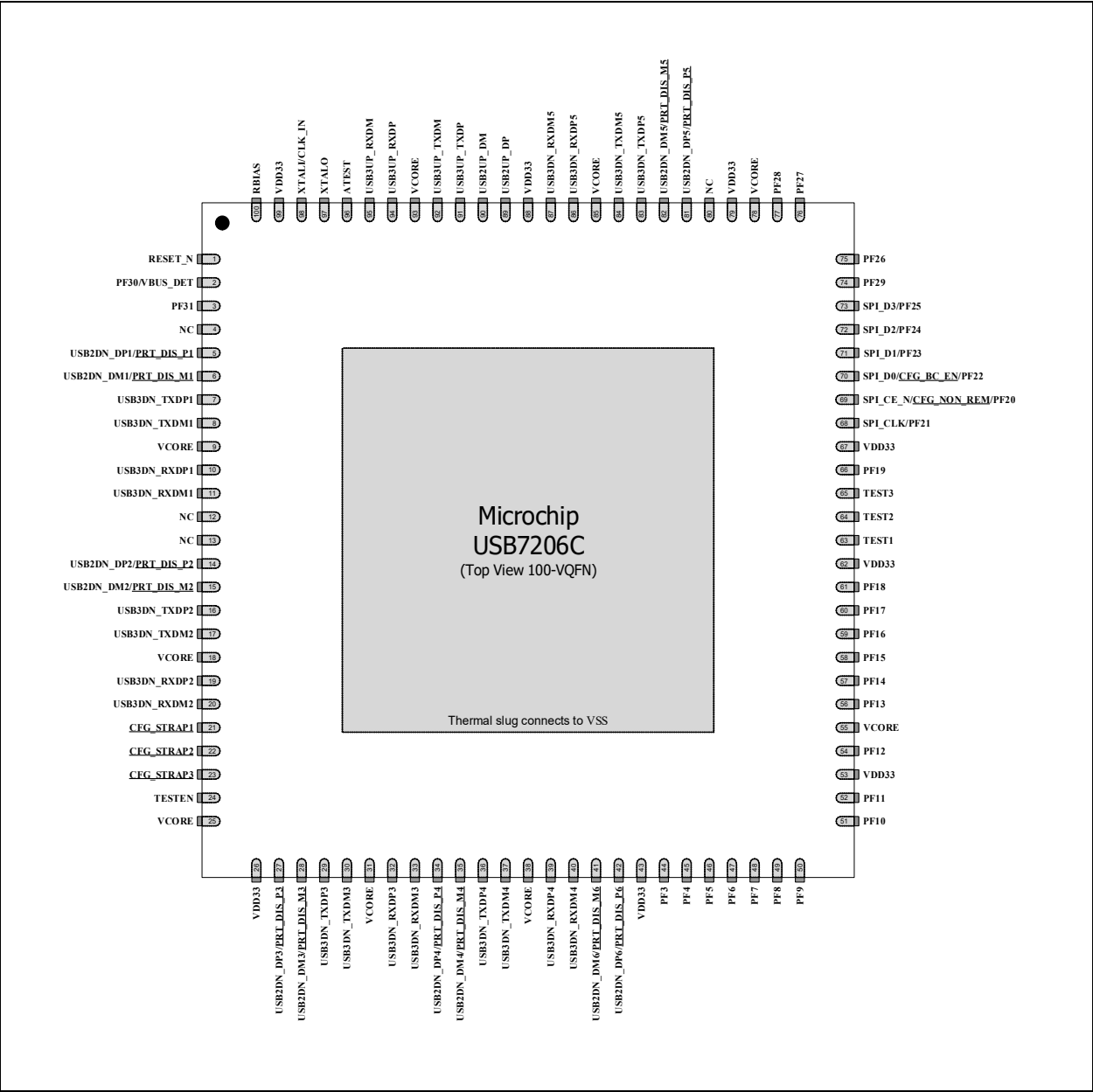
Note: All port numbering in this document is LOGICAL port numbering with the device in the default configuration. LOGICAL port numbering is the numbering as communicated to the USB host. It is the end result after any port number remapping or port disabling. The PHYSICAL port number is the port number with respect to the physical PHY on the chip. PHYSICAL port numbering is fixed and the settings are not impacted by LOGICAL port renumbering/remapping. Certain port settings are made with respect to LOGICAL port numbering, and other port settings are made with respect to PHYSICAL port numbering. Refer to the “Configuration of USB7202/USB7206/USB725x” application note for details on the LOGICAL vs. PHYSICAL mapping and additional configuration details.

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3.0 PIN DESCRIPTIONS

3.1 Pin Assignments

FIGURE 3-1: USB7206C 100-VQFN PIN ASSIGNMENTS



Note: Configuration straps are identified by an underlined symbol name. Signals that function as configuration straps must be augmented with an external resistor when connected to a load.

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Pin Num	Pin Name	Reset	Pin Num	Pin Name	Reset
1	RESET_N	Z	51	PF10	PD
2	PF30/VBUS_DET	Z	52	PF11	PD
3	PF31	Z	53	VDD33	Z
4	NC	AI	54	PF12	PD
5	USB2DN_DP1/PRT_DIS_P1	AIO PD	55	VCORE	Z
6	USB2DN_DM1/PRT_DIS_M1	AIO PD	56	PF13	PD
7	USB3DN_TXDP1	AO PD	57	PF14	PD
8	USB3DN_TXDM1	AO PD	58	PF15	PD
9	VCORE	Z	59	PF16	PD
10	USB3DN_RXDP1	AI PD	60	PF17	PD
11	USB3DN_RXDM1	AI PD	61	PF18	Z
12	NC	AI	62	VDD33	Z
13	NC	AI	63	TEST1	Z
14	USB2DN_DP2/PRT_DIS_P2	AIO PD	64	TEST2	Z
15	USB2DN_DM2/PRT_DIS_M2	AIO PD	65	TEST3	Z
16	USB3DN_TXDP2	AO PD	66	PF19	Z
17	USB3DN_TXDM2	AO PD	67	VDD33	Z
18	VCORE	Z	68	SPI_CLK/PF21	Z
19	USB3DN_RXDP2	AI PD	69	SPI_CE_N/CFG_NON_REM/PF20	PU
20	USB3DN_RXDM2	AI PD	70	SPI_D0/CFG_BC_EN/PF22	Z
21	CFG_STRAP1	Z	71	SPI_D1/PF23	Z
22	CFG_STRAP2	Z	72	SPI_D2/PF24	Z
23	CFG_STRAP3	Z	73	SPI_D3/PF25	Z
24	TESTEN	Z	74	PF29	Z
25	VCORE	Z	75	PF26	Z
26	VDD33	Z	76	PF27	Z
27	USB2DN_DP3/PRT_DIS_P3	AIO PD	77	PF28	Z
28	USB2DN_DM3/PRT_DIS_M3	AIO PD	78	VCORE	Z
29	USB3DN_TXDP3	AO PD	79	VDD33	Z
30	USB3DN_TXDM3	AO PD	80	NC	AI
31	VCORE	Z	81	USB2DN_DP5/PRT_DIS_P5	AIO PD
32	USB3DN_RXDP3	AI PD	82	USB2DN_DM5/PRT_DIS_M5/USB2DN_DP5/ PRT_DIS_P5	AIO PD
33	USB3DN_RXDM3	AI PD	83	USB3DN_TXDP5	AO PD
34	USB2DN_DP4/PRT_DIS_P4	AIO PD	84	USB3DN_TXDM5	AO PD
35	USB2DN_DM4/PRT_DIS_M4	AIO PD	85	VCORE	Z
36	USB3DN_TXDP4	AO PD	86	USB3DN_RXDP5	AI PD
37	USB3DN_TXDM4	AO PD	87	USB3DN_RXDM5	AI PD
38	VCORE	Z	88	VDD33	Z
39	USB3DN_RXDP4	AI PD	89	USB2UP_DP	AIO Z
40	USB3DN_RXDM4	AI PD	90	USB2UP_DM	AIO Z
41	USB2DN_DM6/PRT_DIS_M6	AIO PD	91	USB3UP_TXDP	AO PD
42	USB2DN_DP6/PRT_DIS_P6	AIO PD	92	USB3UP_TXDM	AO PD
43	VDD33	Z	93	VCORE	Z
44	PF3	Z	94	USB3UP_RXDP	AI PD
45	PF4	Z	95	USB3UP_RXDM	AI PD
46	PF5	Z	96	ATEST	AO
47	PF6	Z	97	XTALO	AO
48	PF7	Z	98	XTALI/CLK_IN	AI
49	PF8	Z	99	VDD33	Z
50	PF9	Z	100	RBIAS	AI

Exposed Pad (VSS) must be connected to ground.

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3.2 Pin Descriptions

This section contains descriptions of the various USB7206C pins. The “_N” symbol in the signal name indicates that the active, or asserted, state occurs when the signal is at a low voltage level. For example, **RESET_N** indicates that the reset signal is active low. When “_N” is not present after the signal name, the signal is asserted when at the high voltage level.

The terms assertion and negation are used exclusively. This is done to avoid confusion when working with a mixture of “active low” and “active high” signal. The term assert, or assertion, indicates that a signal is active, independent of whether that level is represented by a high or low voltage. The term negate, or negation, indicates that a signal is inactive.

The “If Unused” column provides information on how to terminate pins if they are unused in a customer design.

Buffer type definitions are detailed in [Section 1.2, Buffer Types](#).

TABLE 3-1: PIN DESCRIPTIONS

Name	Symbol	Buffer Type	Description	If Unused
USB 3.2 Gen 2 Interfaces				
Upstream USB 3.2 Gen 2 TX D+	USB3UP_TXDP	I/O-U	Upstream USB 3.2 Gen 2 Transmit Data Plus.	Float
Upstream USB 3.2 Gen 2 TX D-	USB3UP_TXDM	I/O-U	Upstream USB 3.2 Gen 2 Transmit Data Minus.	Float
Upstream USB 3.2 Gen 2 RX D+	USB3UP_RXDP	I/O-U	Upstream USB 3.2 Gen 2 Receive Data Plus.	Weak pull-down to GND
Upstream USB 3.2 Gen 2 RX D-	USB3UP_RXDM	I/O-U	Upstream USB 3.2 Gen 2 Receive Data Minus.	Weak pull-down to GND
Downstream Ports 1-5 USB 3.2 Gen 2 TX D+	USB3DN_TXDP[1:5]	I/O-U	Downstream SuperSpeed+ Transmit Data Plus, ports 1 through 5.	Float
Downstream Ports 1-5 USB 3.2 Gen 2 TX D-	USB3DN_TXDM[1:5]	I/O-U	Downstream SuperSpeed+ Transmit Data Minus, ports 1 through 5.	Float
Downstream Ports 1-5 USB 3.2 Gen 2 RX D+	USB3DN_RXDP[1:5]	I/O-U	Downstream SuperSpeed+ Receive Data Plus, ports 1 through 5.	Weak pull-down to GND
Downstream Ports 1-5 USB 3.2 Gen 2 RX D-	USB3DN_RXDM[1:5]	I/O-U	Downstream SuperSpeed+ Receive Data Minus, ports 1 through 5.	Weak pull-down to GND
USB 2.0 Interfaces				
Upstream USB 2.0 D+	USB2UP_DP	I/O-U	Upstream USB 2.0 Data Plus (D+).	Mandatory Note 3-6
Upstream USB 2.0 D-	USB2UP_DM	I/O-U	Upstream USB 2.0 Data Minus (D-).	Mandatory Note 3-6
Downstream Ports 1-6 USB 2.0 D+	USB2DN_DP[1:6]	I/O-U	Downstream USB 2.0 Ports 1-6 Data Plus (D+).	Connect directly to 3.3V

TABLE 3-1: PIN DESCRIPTIONS (CONTINUED)

Name	Symbol	Buffer Type	Description	If Unused
Downstream Ports 1-6 USB 2.0 D-	USB2DN_DM[1:6]	I/O-U	Downstream USB 2.0 Ports 1-6 Data Minus (D-)	Connect directly to 3.3V
Detect Upstream VBUS Power	VBUS_DET	I	The Microchip hub monitors VBUS_DET to determine when to assert the internal D+ pull-up resistor (signaling a connect event). When designing a detachable hub, connect this pin to the VBUS power pin of the USB port that is upstream of the hub. For self-powered applications with a permanently attached host, this pin should be pulled up, typically to VDD33. VBUS_DET is a 3.3V input. A resistor divider must be used when connecting to 5V of USB power.	Pull up
SPI Interface				
SPI Clock	SPI_CLK	I/O-U	SPI clock. If the SPI interface is enabled, this pin must be driven low during reset.	Weak pull-down to GND
SPI Data 3-0	SPI_D[3:0]	I/O-U	SPI Data 3-0. If the SPI interface is enabled, these signals function as Data 3 through 0. Note 3-1 SPI_D0 operates as the CFG_BC_EN strap if external SPI memory is not used. It must be terminated with the selected strap resistor to 3.3V or GND. SPI_D[1:3] should be connected to GND through a weak pull-down.	Note 3-1
SPI Chip Enable	SPI_CE_N	I/O12	Active low SPI chip enable input. If the SPI interface is enabled, this pin must be driven high in powerdown states. Note 3-2 Operates as the CFG_NON_REM strap if external SPI memory is not used. It must be terminated with the selected strap resistor to 3.3V or GND.	Note 3-2
Miscellaneous				
Programmable Function Pins	PF[31:3]	I/O12	Programmable function pins. Note 3-3 If unused: depends on the configured pin function. Refer to Section 3.3.4, PF[31:3] Configuration (CFG_STRAP[2:1])	Note 3-3

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TABLE 3-1: PIN DESCRIPTIONS (CONTINUED)

Name	Symbol	Buffer Type	Description	If Unused
Test 1	TEST1	A	Test 1 pin. This signal is used for test purposes and must always be pulled-up to 3.3V via a 10 kΩ resistor.	Pull to 3.3V through a 10 kΩ resistor
Test 2	TEST2	A	Test 2 pin. This signal is used for test purposes and must always be pulled-up to 3.3V via a 10 kΩ resistor.	Pull to 3.3V through a 10 kΩ resistor
Test 3	TEST3	A	Test 3 pin. This signal is used for test purposes and must always be pulled-up to 3.3V via a 10 kΩ resistor.	Pull to 3.3V through a 10 kΩ resistor
Reset Input	RESET_N	IS	This active low signal is used by the system to reset the device.	Mandatory Note 3-6
Bias Resistor	RBIAS	I-R	A 12.0 kΩ ±1.0% resistor is attached from ground to this pin to set the transceiver's internal bias settings. Place the resistor as close the device as possible with a dedicated, low impedance connection to the ground plane.	Mandatory Note 3-6
Test	TESTEN	I/O12	Test pin. This signal is used for test purposes and must always be connected to ground.	Connect to GND
Analog Test	ATEST	A	Analog test pin. This signal is used for test purposes and must always be left unconnected.	Float
External 25 MHz Crystal Input	XTALI	ICLK	External 25 MHz crystal input	Mandatory Note 3-6
External 25 MHz Reference Clock Input	CLK_IN	ICLK	External reference clock input. The device may alternatively be driven by a single-ended clock oscillator. When this method is used, XTALO should be left unconnected.	Mandatory Note 3-6
External 25 MHz Crystal Output	XTALO	OCLK	External 25 MHz crystal output	Float (only if single-ended clock is connected to CLK_IN)

TABLE 3-1: PIN DESCRIPTIONS (CONTINUED)

Name	Symbol	Buffer Type	Description	If Unused
No Connect	NC	-	No connect. For proper operation, this pin must be left unconnected.	No connect
Configuration Straps				
Port 6-1 D+ Disable Configuration Strap	<u>PRT_DIS_P[6:1]</u>	I	Port 6-1 D+ Disable Configuration Strap. These configuration straps are used in conjunction with the corresponding <u>PRT_DIS_M[6:1]</u> straps to disable the related port (6-1). See Note 3-7 . Both USB data pins for the corresponding port must be tied to 3.3V to disable the associated downstream port.	N/A
Port 6-1 D- Disable Configuration Strap	<u>PRT_DIS_M[6:1]</u>	I	Port 6-1 D- Disable Configuration Strap. These configuration straps are used in conjunction with the corresponding <u>PRT_DIS_P[6:1]</u> straps to disable the related port (6-1). See Note 3-7 . Both USB data pins for the corresponding port must be tied to 3.3V to disable the associated downstream port.	Mandatory Note 3-6
Non-Removable Ports Configuration Strap	<u>CFG_NON_REM</u>	I	Non-Removable Ports Configuration Strap. This configuration strap controls the number of reported non-removable ports. See Note 3-7 . Note 3-4 Mandatory if external SPI memory is not used for firmware execution. If external SPI memory is used for firmware execution, then configuration strap resistor should be omitted.	Note 3-4

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TABLE 3-1: PIN DESCRIPTIONS (CONTINUED)

Name	Symbol	Buffer Type	Description	If Unused
Battery Charging Configuration Strap	<u>CFG_BC_EN</u>	I/O12	Battery Charging Configuration Strap. This configuration strap controls the number of BC 1.2 enabled downstream ports. See Note 3-7. Note 3-5 Mandatory if external SPI memory is not used for firmware execution. If external SPI memory is used for firmware execution, then configuration strap resistor should be omitted.	Mandatory Note 3-6
Device Mode Configuration Straps 3-1	<u>CFG_STRAP[3:1]</u>	I	Device Mode Configuration Straps 3-1. These configuration straps are used to select the device's mode of operation. See Note 3-7.	Mandatory Note 3-6
Power/Ground				
+3.3V I/O Power Supply Input	VDD33	P	+3.3 V power supply input.	Mandatory Note 3-7
Digital Core Power Supply Input	VCORE	P	Digital core power supply input.	Mandatory Note 3-7
Ground	VSS	P	Common ground. The exposed pad must be connected to the ground plane with a via array.	Mandatory Note 3-7

Note 3-6 Configuration strap values are latched on Power-On Reset (POR) and the rising edge of **RESET_N** (external chip reset). Configuration straps are identified by an underlined symbol name. Signals that function as configuration straps must be augmented with an external resistor when connected to a load. For additional information, refer to [Section 3.3, Configuration Straps and Programmable Functions](#).

Note 3-7 Pin use is mandatory. Cannot be left unused.

3.3 Configuration Straps and Programmable Functions

Configuration straps are multi-function pins that are used during Power-On Reset (POR) or external chip reset (**RESET_N**) to determine the default configuration of a particular feature. The state of the signal is latched following deassertion of the reset. Configuration straps are identified by an underlined symbol name. This section details the various device configuration straps and associated programmable pin functions.

Note: The system designer must guarantee that configuration straps meet the timing requirements specified in [Section 9.6.2, Power-On and Configuration Strap Timing](#) and [Section 9.6.3, Reset and Configuration Strap Timing](#). If configuration straps are not at the correct voltage level prior to being latched, the device may capture incorrect strap values.

3.3.1 PORT DISABLE CONFIGURATION (PRT_DIS_P[6:1] / PRT_DIS_M[6:1])

The PRT_DIS_P[6:1] / PRT_DIS_M[6:1] configuration straps are used in conjunction to disable the related port (6-1).

For PRT_DIS_P_x (where *x* is the corresponding port 6-1):

0 = Port *x* D+ Enabled

1 = Port *x* D+ Disabled

For PRT_DIS_M_x (where *x* is the corresponding port 6-1):

0 = Port *x* D- Enabled

1 = Port *x* D- Disabled

3.3.2 NON-REMOVABLE PORT CONFIGURATION (CFG_NON_REM)

Note: Both PRT_DIS_P_x and PRT_DIS_M_x (where *x* is the corresponding port) must be tied to 3.3V to disable the associated downstream port. Disabling the USB 2.0 port will also disable the corresponding USB 3.0 port.

The CFG_NON_REM configuration strap is used to configure the non-removable port settings of the device to one of six settings. These modes are selected by the configuration of an external resistor on the CFG_NON_REM pin. The resistor options are a 200 kΩ pull-down, 200 kΩ pull-up, 10 kΩ pull-down, 10 kΩ pull-up, 10 Ω pull-down, and 10 Ω pull-up, as shown in [Table 3-2](#).

TABLE 3-2: CFG_NON_REM RESISTOR ENCODING

<u>CFG_NON_REM</u> Resistor Value	Setting
200 kΩ Pull-Down	All ports removable
200 kΩ Pull-Up	Port 1 non-removable
10 kΩ Pull-Down	Ports 1, 2 non-removable
10 kΩ Pull-Up	Ports 1, 2, 3 non-removable
10 Ω Pull-Down	Ports 1, 2, 3, 4 non-removable
10 Ω Pull-Up	Ports 1, 2, 3, 4, 5, 6 non-removable

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3.3.3 BATTERY CHARGING CONFIGURATION (CFG_BC_EN)

The CFG_BC_EN configuration strap is used to configure the battery charging port settings of the device to one of six settings. These modes are selected by the configuration of an external resistor on the CFG_BC_EN pin. The resistor options are a 200 k Ω pull-down, 200 k Ω pull-up, 10 k Ω pull-down, 10 k Ω pull-up, 10 Ω pull-down, and 10 Ω pull-up, as shown in [Table 3-3](#).

TABLE 3-3: CFG_BC_EN RESISTOR ENCODING

<u>CFG_BC_EN</u> Resistor Value	Setting
200 k Ω Pull-Down	Battery charging not enable on any port
200 k Ω Pull-Up	BC1.2 DCP and CDP battery charging enabled on Port 1
10 k Ω Pull-Down	BC1.2 DCP and CDP battery charging enabled on Ports 1, 2
10 k Ω Pull-Up	BC1.2 DCP and CDP battery charging enabled on Ports 1, 2, 3
10 Ω Pull-Down	BC1.2 DCP and CDP battery charging enabled on Ports 1, 2, 3, 4
10 Ω Pull-Up	BC1.2 DCP and CDP battery charging enabled on Ports 1, 2, 3, 4, 5, 6

3.3.4 PF[31:3] CONFIGURATION (CFG_STRAP[2:1])

The USB7206C provides 29 programmable function pins (PF[31:3]). These pins can only be configured to 1 predefined configuration via the CFG_STRAP[2:1] pins. This configuration is selected via external resistors on the CFG_STRAP[2:1] pins, as detailed in [Table 3-4](#). Resistor values and combinations not detailed in [Table 3-4](#) are reserved and should not be used.

Note: CFG_STRAP3 is not used and must be pulled-down to ground via a 200 k Ω resistor.

TABLE 3-4: CFG_STRAP[2:1] RESISTOR ENCODING

Mode	<u>CFG_STRAP2</u> Resistor Value	<u>CFG_STRAP1</u> Resistor Value
Configuration 3	200 k Ω Pull-Down	10 k Ω Pull-Down

Note: Configurations 1 and 2 are not used in the USB7206C.

A summary of the configuration pin assignments is provided in [Table 3-5](#). For details on behavior of each programmable function, refer to [Table 3-6](#).

TABLE 3-5: PF[31:3] FUNCTION ASSIGNMENT

Pin	Configuration 3
PF3	I2S_SDI
PF4	I2S_SDO
PF5	I2S_SCK
PF6	I2S_LRCK
PF7	I2S_MCLK
PF8	NC
PF9	NC
PF10	PRT_CTL3_U3
PF11	PRT_CTL4_U3
PF12	PRT_CTL5_U3
PF13	PRT_CTL5
PF14	PRT_CTL4
PF15	PRT_CTL3
PF16	PRT_CTL2
PF17	PRT_CTL1
PF18	MSTR_I2C_CLK
PF19	MIC_DET
PF20	SPI_CE_N
PF21	SPI_CLK
PF22	SPI_D0
PF23	SPI_D1
PF24	SPI_D2
PF25	SPI_D3
PF26	SLV_I2C_CLK
PF27	SLV_I2C_DATA
PF28	PRT_CTL6
PF29	GPIO93
PF30	VBUS_DET
PF31	MSTR_I2C_DATA

Note: The default **PF_x** pin functions can be overridden with additional configuration by modification of the pin mux registers. These changes can be made during the SMBus configuration stage, by programming to OTP memory, or during runtime (after hub has attached and enumerated) by register writes via the SMBus slave interface or USB commands to the internal Hub Feature Controller Device.

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TABLE 3-6: PROGRAMMABLE FUNCTIONS DESCRIPTIONS

Function	Buffer Type	Description	If Unused
Master SMBus/I²C Interface			
MSTR_I2C_CLK	I/O12	Bridging Master SMBus/I ² C controller clock (SMBus/I ² C controller 1). External 1k-10k pull-up resistors to 3.3V are required if the I ² C Master Interface is to be used.	Weak pull-down to GND
MSTR_I2C_DATA	I/O12	Bridging Master SMBus/I ² C controller data (SMBus/I ² C controller 1). External 1k-10k pull-up resistors to 3.3V are required if the I ² C Master Interface is to be used.	Weak pull-down to GND
Slave SMBus/I²C Interface			
SLV_I2C_CLK	I/O12	Slave SMBus/I ² C controller clock (SMBus/I ² C controller 2). External 1k-10k pull-up resistors to 3.3V are required if the I ² C Slave Interface is to be used.	Weak pull-down to GND
SLV_I2C_DATA	I/O12	Slave SMBus/I ² C controller data (SMBus/I ² C controller 2). External 1k-10k pull-up resistors to 3.3V are required if the I ² C Slave Interface is to be used.	Weak pull-down to GND
I²S Interface			
I2S_SDI	I	I ² S Serial Data In	Weak pull-down to GND
I2S_SDO	O12	I ² S Serial Data Out	Weak pull-down to GND
I2S_SCK	O12	I ² S Continuous Serial Clock	Weak pull-down to GND
I2S_LRCK	O12	I ² S Word Select / Left-Right Clock	Weak pull-down to GND
I2S_MCLK	O12	I ² S Master Clock	Weak pull-down to GND
MIC_DET	I	I ² S Microphone Plug Detect 0 = No microphone plugged into the audio jack 1 = Microphone plugged into the audio jack	Weak pull-down to GND

TABLE 3-6: PROGRAMMABLE FUNCTIONS DESCRIPTIONS (CONTINUED)

Function	Buffer Type	Description	If Unused
Miscellaneous			
PRT_CTL6	I/O12 (PU)	Port 6 power enable / overcurrent sense When the downstream port is enabled, this pin is set as an input with an internal pull-up resistor applied. The internal pull-up enables power to the downstream port while the pin monitors for an active low overcurrent signal assertion from an external current monitor on USB port 6. This pin will change to an output and be driven low when the port is disabled by configuration or by the host control. Note: This signal controls both the USB 2.0 and USB 3.2 portions of the port. Note 3-1 This pin can be left unused only if Port 6 is disabled via strap/OTP/SMBus/SPI configuration.	Float (Note 3-1)
PRT_CTL5	I/O12 (PU)	Port 5 power enable / overcurrent sense When the downstream port is enabled, this pin is set as an input with an internal pull-up resistor applied. The internal pull-up enables power to the downstream port while the pin monitors for an active low overcurrent signal assertion from an external current monitor on USB port 5. This pin will change to an output and be driven low when the port is disabled by configuration or by the host control. Note: When PortSplit is disabled, this signal controls both the USB 2.0 and USB 3.2 portions of the port. When PortSplit is enabled, this signal controls the USB 2.0 portion of the port only. Note 3-2 This pin can be left unused only if Port 5 is disabled via strap/OTP/SMBus/SPI configuration.	Float (Note 3-2)
PRT_CTL4	I/O12 (PU)	Port 4 power enable / overcurrent sense When the downstream port is enabled, this pin is set as an input with an internal pull-up resistor applied. The internal pull-up enables power to the downstream port while the pin monitors for an active low overcurrent signal assertion from an external current monitor on USB port 4. This pin will change to an output and be driven low when the port is disabled by configuration or by the host control. Note: When PortSplit is disabled, this signal controls both the USB 2.0 and USB 3.2 portions of the port. When PortSplit is enabled, this signal controls the USB 2.0 portion of the port only. Note 3-3 This pin can be left unused only if Port 4 is disabled via strap/OTP/SMBus/SPI configuration.	Float (Note 3-3)

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TABLE 3-6: PROGRAMMABLE FUNCTIONS DESCRIPTIONS (CONTINUED)

Function	Buffer Type	Description	If Unused
PRT_CTL3	I/O12 (PU)	Port 3 power enable / overcurrent sense When the downstream port is enabled, this pin is set as an input with an internal pull-up resistor applied. The internal pull-up enables power to the downstream port while the pin monitors for an active low overcurrent signal assertion from an external current monitor on USB port 3. This pin will change to an output and be driven low when the port is disabled by configuration or by the host control. Note: When PortSplit is disabled, this signal controls both the USB 2.0 and USB 3.2 portions of the port. When PortSplit is enabled, this signal controls the USB 2.0 portion of the port only. Note 3-4 This pin can be left unused only if Port 3 is disabled via strap/OTP/SMBus/SPI configuration.	Float (Note 3-4)
PRT_CTL2	I/O12 (PU)	Port 2 power enable / overcurrent sense When the downstream port is enabled, this pin is set as an input with an internal pull-up resistor applied. The internal pull-up enables power to the downstream port while the pin monitors for an active low overcurrent signal assertion from an external current monitor on USB port 2. This pin will change to an output and be driven low when the port is disabled by configuration or by the host control. Note: When PortSplit is disabled, this signal controls both the USB 2.0 and USB 3.2 portions of the port. When PortSplit is enabled, this signal controls the USB 2.0 portion of the port only. Note 3-5 This pin can be left unused only if Port 2 is disabled via strap/OTP/SMBus/SPI configuration.	Float (Note 3-1)
PRT_CTL1	I/O12 (PU)	Port 1 power enable / overcurrent sense When the downstream port is enabled, this pin is set as an input with an internal pull-up resistor applied. The internal pull-up enables power to the downstream port while the pin monitors for an active low overcurrent signal assertion from an external current monitor on USB port 1. This pin will change to an output and be driven low when the port is disabled by configuration or by the host control. Note: When PortSplit is disabled, this signal controls both the USB 2.0 and USB 3.2 portions of the port. When PortSplit is enabled, this signal controls the USB 2.0 portion of the port only. Note 3-6 This pin can be left unused only if Port 1 is disabled via strap/OTP/SMBus/SPI configuration.	Float (Note 3-1)

TABLE 3-6: PROGRAMMABLE FUNCTIONS DESCRIPTIONS (CONTINUED)

Function	Buffer Type	Description	If Unused
PRT_CTL5_U3	O12	Port 5 USB 3.2 PortSplit power enable This signal is an active high control signal used to enable to the USB 3.2 portion of the downstream port 5 when PortSplit is enabled. When PortSplit is disabled, this pin is not used. Note: This signal should only be used to control an embedded USB 3.2 device.	Float
PRT_CTL4_U3	O12	Port 4 USB 3.2 PortSplit power enable This signal is an active high control signal used to enable to the USB 3.2 portion of the downstream port 4 when PortSplit is enabled. When PortSplit is disabled, this pin is not used. Note: This signal should only be used to control an embedded USB 3.2 device.	Float
PRT_CTL3_U3	O12	Port 3 USB 3.2 PortSplit power enable This signal is an active high control signal used to enable to the USB 3.2 portion of the downstream port 3 when PortSplit is enabled. When PortSplit is disabled, this pin is not used. Note: This signal should only be used to control an embedded USB 3.2 device.	Float

3.4 Physical and Logical Port Mapping

The USB72xx family of devices are based upon a common architecture, but all have different modifications and/or pin bond outs to achieve the various device configurations. The base chip is composed of a total of 6 USB3 PHYs and 7 USB2 PHYs. These PHYs are physically arranged on the chip in a certain way, which is referred to as the PHYSICAL port mapping.

The actual port numbering is remapped by default in different ways on each device in the family. This changes the way that the ports are numbered from the USB host's perspective. This is referred to as LOGICAL mapping.

The various configuration options available for these devices may, at times, be with respect to PHYSICAL mapping or LOGICAL mapping. Each individual configuration option which has a PHYSICAL or LOGICAL dependency is declared as such within the register description.

The PHYSICAL vs. LOGICAL mapping is described for all port related pins in [Table 3-7](#). A system design in schematics and layout is generally performed using the pinout in [Section 3.1, Pin Assignments](#), which is assigned by the default LOGICAL mapping. Hence, it may be necessary to cross reference the PHYSICAL vs. LOGICAL look up tables when determining the hub configuration.

Note: The MPLAB Connect tool makes configuration simple; the settings can be selected by the user with respect to the LOGICAL port numbering. The tool handles the necessary linking to the PHYSICAL port settings. Refer to [Section 6.0, Device Configuration](#) for additional information.

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TABLE 3-7: USB7206C PHYSICAL VS. LOGICAL PORT MAPPING

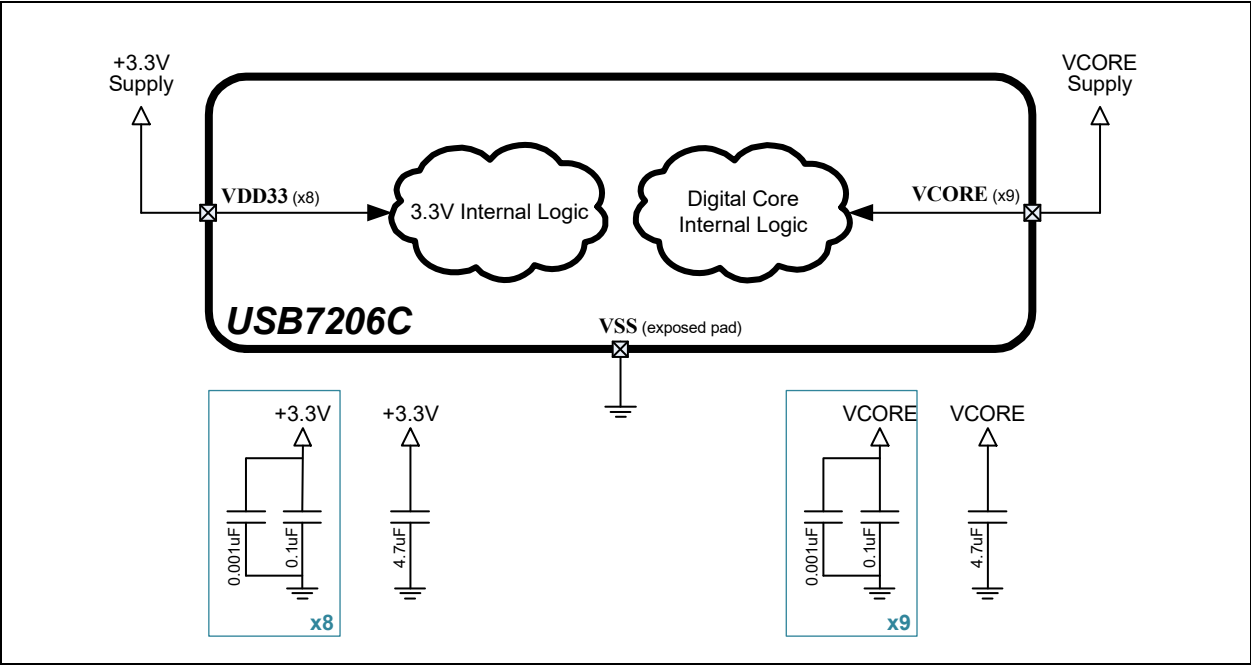
Device Pin	Pin Name (as in datasheet)	LOGICAL PORT NUMBER							PHYSICAL PORT NUMBER						
		0	1	2	3	4	5	6	0	1	2	3	4	5	6
5	USB2DN_DP1		X							X					
6	USB2DN_DM1		X							X					
7	USB3DN_TXDP1		X							X					
8	USB3DN_TXDM1		X							X					
10	USB3DN_RXDP1		X							X					
11	USB3DN_RXDM1		X							X					
14	USB2DN_DP2			X							X				
15	USB2DN_DM2			X							X				
16	USB3DN_TXDP2			X							X				
17	USB3DN_TXDM2			X							X				
19	USB3DN_RXDP2			X							X				
20	USB3DN_RXDM2			X							X				
27	USB2DN_DP3				X							X			
28	USB2DN_DM3				X							X			
29	USB3DN_TXDP3				X							X			
30	USB3DN_TXDM3				X							X			
32	USB3DN_RXDP3				X							X			
33	USB3DN_RXDM3				X							X			
34	USB2DN_DP4					X							X		
35	USB2DN_DM4					X							X		
36	USB3DN_TXDP4					X							X		
37	USB3DN_TXDM4					X							X		
39	USB3DN_RXDP4					X							X		
40	USB3DN_RXDM4					X							X		
41	USB2DN_DM6							X							X
42	USB2DN_DP6							X							X
81	USB2DN_DP5						X							X	
82	USB2DN_DM5						X							X	
83	USB3DN_TXDP5						X							X	
84	USB3DN_TXDM5						X							X	
86	USB3DN_RXDP5						X							X	
87	USB3DN_RXDM5						X							X	
89	USB2UP_DP	X							X						
90	USB2UP_DM	X							X						
91	USB3UP_TXDP	X							X						
92	USB3UP_TXDM	X							X						
94	USB3UP_RXDP	X							X						
95	USB3UP_RXDM	X							X						

4.0 DEVICE CONNECTIONS

4.1 Power Connections

Figure 4-1 illustrates the device power connections.

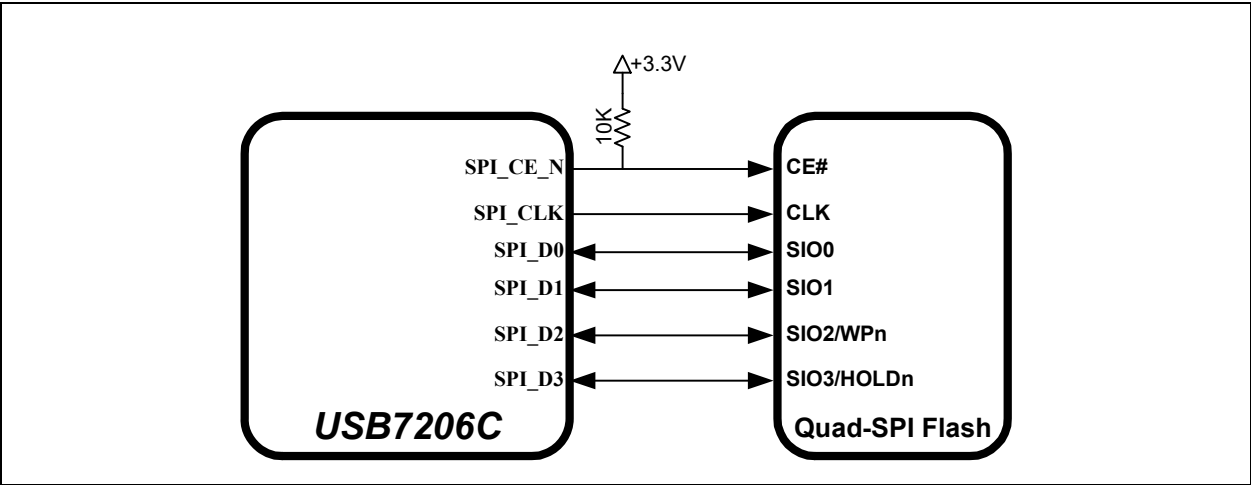
FIGURE 4-1: POWER CONNECTIONS



4.2 SPI Flash Connections

Figure 4-2 illustrates the Quad-SPI flash connections.

FIGURE 4-2: QUAD-SPI FLASH CONNECTIONS

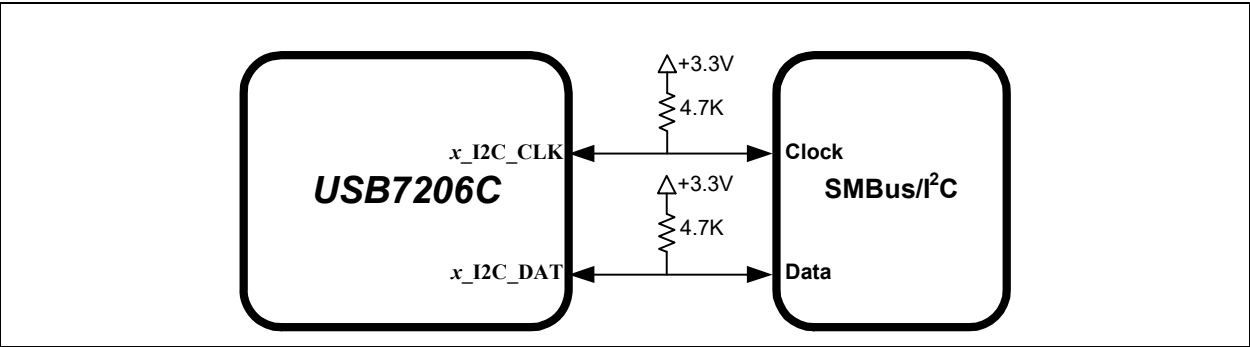


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4.3 SMBus/I²C Connections

Figure 4-3 illustrates the SMBus/I²C connections.

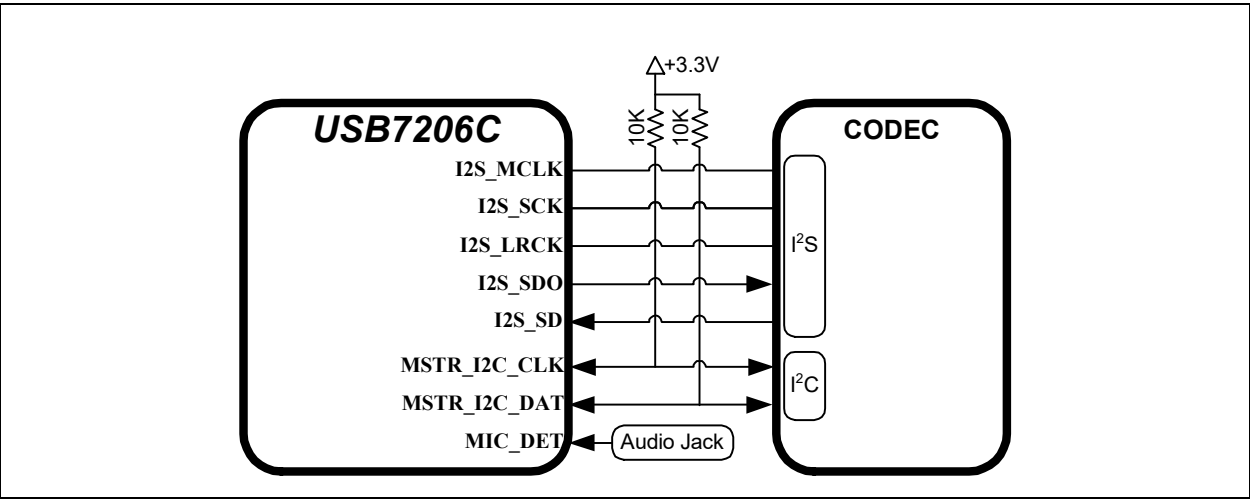
FIGURE 4-3: SMBUS/I²C CONNECTIONS



4.4 I²S Connections

Figure 4-4 illustrates the I²S connections.

FIGURE 4-4: I²S CONNECTIONS



5.0 MODES OF OPERATION

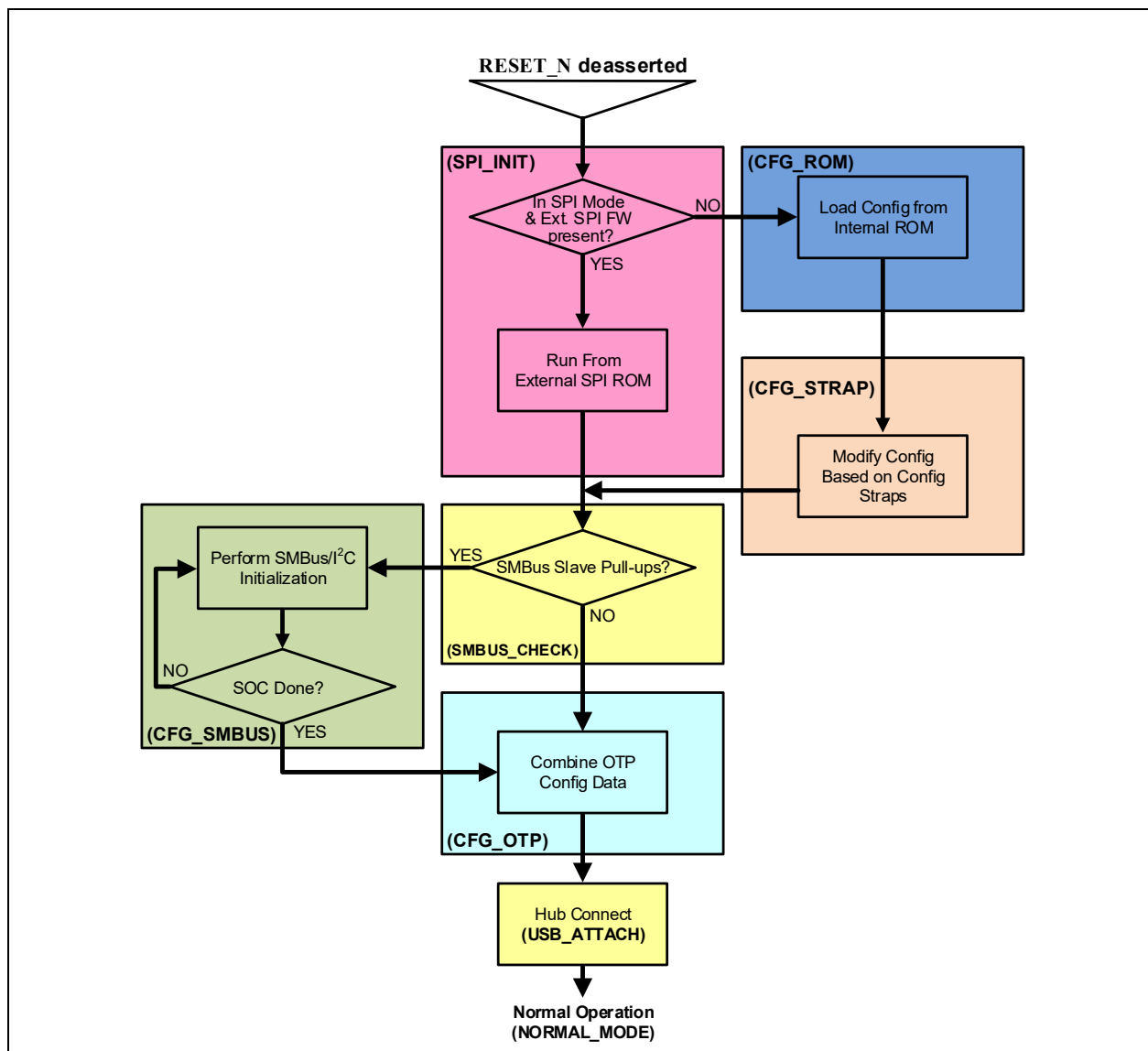
The device provides two main modes of operation: Standby Mode and Hub Mode. These modes are controlled via the **RESET_N** pin, as shown in [Table 5-1](#).

TABLE 5-1: MODES OF OPERATION

RESET_N Input	Summary
0	Standby Mode: This is the lowest power mode of the device. No functions are active other than monitoring the RESET_N input. All port interfaces are high impedance and the PLL is halted. Refer to Section 8.9, Resets for additional information on RESET_N .
1	Hub (Normal) Mode: The device operates as a configurable USB hub. This mode has various sub-modes of operation, as detailed in Figure 5-1 . Power consumption is based on the number of active ports, their speed, and amount of data received.

The flowchart in [Figure 5-1](#) details the modes of operation and details how the device traverses through the Hub Mode stages (shown in bold). The remaining sub-sections provide more detail on each stage of operation.

FIGURE 5-1: HUB MODE FLOWCHART



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5.1 Boot Sequence

5.1.1 STANDBY MODE

If the **RESET_N** pin is asserted, the hub will be in Standby Mode. This mode provides a very low power state for maximum power efficiency when no signaling is required. This is the lowest power state. In Standby Mode all downstream ports are disabled, the USB data pins are held in a high-impedance state, all transactions immediately terminate (no states saved), all internal registers return to their default state, the PLL is halted, and core logic is powered down in order to minimize power consumption. Because core logic is powered off, no configuration settings are retained in this mode and must be re-initialized after **RESET_N** is negated high.

5.1.2 SPI INITIALIZATION STAGE (SPI_INIT)

The first stage, the initialization stage, occurs on the deassertion of **RESET_N**. In this stage, the internal logic is reset, the PLL locks if a valid clock is supplied, and the configuration registers are initialized to their default state. The internal firmware then checks for an external SPI ROM. The firmware looks for an external SPI flash device that contains a valid signature of "2DFU" (device firmware upgrade) beginning at address 0x3FFFA. If a valid signature is found, then the external SPI ROM is enabled and the code execution begins at address 0x0000 in the external SPI device. If a valid signature is not found, then execution continues from internal ROM (CFG_ROM stage).

The required SPI ROM must be a minimum of 4 Mbit, and 60 MHz or faster. Both 1, 2, and 4-bit SPI operation is supported. For optimum throughput, a 2-bit SPI ROM is recommended. Both mode 0 and mode 3 SPI ROMs are also supported.

If the system is not strapped for SPI Mode, code execution will continue from internal ROM (CFG_ROM stage).

5.1.3 CONFIGURATION FROM INTERNAL ROM STAGE (CFG_ROM)

In this stage, the internal firmware loads the default values from the internal ROM. Most of the hub configuration registers, USB descriptors, electrical settings, etc. will be initialized in this state.

5.1.4 CONFIGURATION STRAP READ STAGE (CFG_STRAP)

In this stage, the firmware reads the following configuration straps to override the default values:

- **CFG_STRAP[3:1]**
- **PRT_DIS_P[6:1]**
- **PRT_DIS_M[6:1]**
- **CFG_NON_REM**
- **CFG_BC_EN**

If the **CFG_STRAP[3:1]** pins are set to Configuration 1, the device will move to the SMBUS_CHECK stage, otherwise it will move to the CFG_OTP stage. Refer to [Section 3.3, Configuration Straps and Programmable Functions](#) for information on usage of the various device configuration straps.

5.1.5 SMBUS CHECK STAGE (SMBUS_CHECK)

Based on the **PF[31:3]** configuration selected (refer to [Section 3.3.4, PF\[31:3\] Configuration \(CFG_STRAP\[2:1\]\)](#)), the firmware will check for the presence of external pull up resistors on the SMBus slave programmable function pins. If 10K pull-ups are detected on both pins, the device will be configured as an SMBus slave, and the next state will be CFG_SMBUS. If a pull-up is not detected in either of the pins, the next state is CFG_OTP.

5.1.6 SMBUS CONFIGURATION STAGE (CFG_SMBUS)

In this stage, the external SMBus master can modify any of the default configuration settings specified in the integrated ROM, such as USB device descriptors, port electrical settings, and control features such as downstream battery charging.

There is no time limit on this mode. In this stage the firmware will wait indefinitely for the SMBus/I²C configuration. The external SMBus master writes to register 0xFF to end the configuration in legacy mode. In non-legacy mode, the SMBus command USB_ATTACH (opcode 0xAA55) or USB_ATTACH_WITH_SMBUS (opcode 0xAA56) will finish the configuration.

5.1.7 OTP CONFIGURATION STAGE (CFG_OTP)

All configuration data from CFG_ROM, CFG_STRAP, CFG_SMBUS stages, and OTP or Pseudo OTP (when SPI firmware is used) are combined in this stage. If a register is modified in both the CFG_SMBUS stage and OTP/Pseudo OTP, the OTP data overwrites SMBus data when executing from ROM firmware, while SMBus data overwrites Pseudo OTP data when executing from SPI firmware. Note that when executing from SPI firmware, the internal OTP content is ignored except for USB2/USB3 UUID data and trim registers for PHY settings.

Note: If the same register is modified in both CFG_SMBUS and CFG_OTP stages, the value from CFG_OTP will overwrite any value written during CFG_SMBUS.

5.1.8 HUB CONNECT STAGE (USB_ATTACH)

After updating the hub registers in CFG_OTP stage, the firmware enables the USB host attachment by setting the USB_ATTACH bit in the HUB_CMD_STAT register for USB 2.0 and the USB3_HUB_ENABLE bit for USB 3.2. The device remains in this USB_ATTACH stage indefinitely until it gets enumerated by a USB Host.

5.1.9 NORMAL MODE (NORMAL_MODE)

Lastly, the hub enters Normal Mode of operation. In this stage full USB operation is supported under control of the USB Host on the upstream port. The device will remain in the normal mode until the operating mode is changed by the system.

If RESET_N is asserted low, then Standby Mode is entered. The device may then be placed into any of the designated hub stages. Asserting a soft disconnect on the upstream port will cause the hub to return to the Hub Connect stage until the soft disconnect is negated.

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6.0 DEVICE CONFIGURATION

The device supports a large number of features (some mutually exclusive), and must be configured in order to correctly function when attached to a USB host controller. Microchip provides a comprehensive software programming tool, MPLAB Connect Configurator (formerly ProTouch2), for OTP configuration of various USB7206C functions and registers. All configuration is to be performed via the MPLAB Connect Configurator programming tool. For additional information on this tool, refer to the MPLAB Connect Configurator programming tool product page at <http://www.microchip.com/design-centers/usb/mplab-connect-configurator>.

Additional information on configuring the USB7206C is also provided in the “*Configuration of the USB720x/USB725x*” application note, which contains details on the hub operational mode, SOC configuration stage, OTP configuration, USB configuration, and configuration register definitions. This application note, along with additional USB7206C resources, can be found on the Microchip USB7206C product page at www.microchip.com/USB7206C.

Note: Device configuration straps and programmable pins are detailed in Section 3.3, Configuration Straps and Programmable Functions. Refer to Section 7.0, Device Interfaces for detailed information on each device interface.
--

7.0 DEVICE INTERFACES

The USB7206C provides multiple interfaces for configuration, external memory access, etc. This section details the various device interfaces:

- [SPI/SQI Master Interface](#)
- [SMBus/I2C Master/Slave Interfaces](#)
- [I2S Interface](#)

For details on how to enable each interface, refer to [Section 3.3, Configuration Straps and Programmable Functions](#).

For information on device connections, refer to [Section 4.0, Device Connections](#). For information on device configuration, refer to [Section 6.0, Device Configuration](#).

Microchip provides a comprehensive software programming tool, MPLAB Connect Configurator (formerly ProTouch2), for configuring the USB7206C functions, registers and OTP memory. All configuration is to be performed via the MPLAB Connect Configurator programming tool. For additional information on this tool, refer to the MPLAB Connect Configurator programming tool product page at <http://www.microchip.com/design-centers/usb/mplab-connect-configurator>.

7.1 SPI/SQI Master Interface

The SPI/SQI controller has two basic modes of operation: execution of an external hub firmware image, or the USB to SPI bridge. On power up, the firmware looks for an external SPI flash device that contains a valid signature of 2DFU (device firmware upgrade) beginning at address 0x3FFFA. If a valid signature is found, then the external ROM mode is enabled and the code execution begins at address 0x0000 in the external SPI device. If a valid signature is not found, then execution continues from internal ROM and the SPI interface can be used as a USB to SPI bridge.

The entire firmware image is then executed in place entirely from the SPI interface. The SPI interface will remain continuously active while the hub is in the runtime state. The hub configuration options are also loaded entirely out of the SPI memory device. Both the internal ROM firmware image and internal OTP memory are completely ignored while executing the firmware and configuration from the external SPI memory.

The second mode of operation is the USB to SPI bridge operation. Additional details on this feature can be found in [Section 8.7, USB to SPI Bridging](#).

[Table 7-1](#) details how the associated pins are mapped in SPI vs. SQI mode.

TABLE 7-1: SPI/SQI PIN USAGE

SPI Mode	SQI Mode	Description
SPI_CE_N	SQI_CE_N	SPI/SQI Chip Enable (Active Low)
SPI_CLK	SQI_CLK	SPI/SQI Clock
SPI_D0	SQI_D0	SPI Data Out; SQI Data I/O 0
SPI_D1	SQI_D1	SPI Data In; SQI Data I/O 1
—	SQI_D2	SQI Data I/O 2
—	SQI_D3	SQI Data I/O 3

Note: For SPI/SQI master timing information, refer to [Section 9.6.10, SPI/SQI Master Timing](#).

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7.2 SMBus/I²C Master/Slave Interfaces

The device provides two independent SMBus/I²C controllers (Slave and Master) which can be used to access internal device run time registers or program the internal OTP memory. The device contains two 128 byte buffers to enable simultaneous master/slave operation and to minimize firmware overhead in processed I²C packets. The I²C interfaces support 100 kHz Standard-mode (Sm) and 400 kHz Fast Mode (Fm) operation.

The SMBus/I²C interfaces are assigned to programmable pins (PF_x). Refer to [Section 3.3.4, PF\[31:3\] Configuration \(CFG_STRAP\[2:1\]\)](#) for additional information.

Note: For SMBus/I²C timing information, refer to [Section 9.6.7, SMBus Timing](#) and [Section 9.6.8, I2C Timing](#).

7.3 I²S Interface

The device provides an integrated I²S interface to facilitate the connection of digital audio devices. The I²S interface conforms to the voltage, power and timing characteristics/specifications as set forth in the *I²S-Bus Specification*, and consists of the following signals:

- **I2S_SDI:** Serial Data Input
- **I2S_SDO:** Serial Data Output
- **I2S_SCK:** Serial Clock
- **I2S_LRCK:** Left/Right Clock (SS/FSYNC)
- **I2S_MCLK:** Master Clock
- **MIC_DET:** Microphone Plug Detect

Each audio connection is half-duplex, so **I2S_SDO** exists only on the transmit side and **I2S_SDI** exists only on the receive side of the interface. Some codecs refer to the Serial Clock (**I2S_SCK**) as Baud/Bit Clock (BCLK). Also, the Left/Right Clock is commonly referred to as LRC or LRCK. The I²S and other audio protocols refer to LRC as Word Select (WS).

The following codec is supported by default:

- Analog Devices ADAU1961 (24-bit 96 kHz)

The I²S interface is assigned to programmable pins (PF_x). Refer to [Section 3.3.4, PF\[31:3\] Configuration \(CFG_STRAP\[2:1\]\)](#) for additional information.

Note: For I²S timing information, refer to [Section 9.6.9, I2S Timing](#). For detailed information on utilizing the I²S interface, including support for other codecs, refer to the application note "*USB720x/USB725x I²S Operation*", which can be found on the Microchip USB7206C product page at www.microchip.com/USB7206C.

7.3.1 MODES OF OPERATION

The USB audio class operates in three ways: Asynchronous, Synchronous and Adaptive. There are also multiple operating modes, such as hi-res, streaming, etc. Typically for USB devices, inputs such as microphones are Asynchronous, and output devices such as speakers are Adaptive. The hardware is set up to handle all three modes of operation. It is recommended that the following configuration be used: Asynchronous IN; Adaptive OUT; 48 kHz streaming mode; Two channels: 16 bits per channel.

7.3.1.1 Asynchronous IN 48 kHz Streaming

In this mode, the codec sampling clock is set to 48 kHz based on the local oscillator. This clock is never changed. The data from the codec is fed into the input FIFO. Since the sampling clock is asynchronous to the host clock, the amount of data captured in every USB frame will vary. This issue is left for the host to handle. The input FIFO has two markers, a low water mark (THRESHOLD_LOW_VAL), and a high water mark (THRESHOLD_HIGH_VAL). There are three registers to determine how much data to send back in each frame. If the amount of data in the FIFO exceeds the high water mark, then HI_PKT_SIZE worth of data is sent. If the data is between the high and low water mark, the normal MID_PKT_SIZE amount of data is sent. If the data is below the low water mark, LO_PKT_SIZE worth of data is sent.

7.3.1.2 Adaptive OUT 48 kHz Streaming

In this mode, the codec sampling clock is initially set to 48 kHz based on the local oscillator. The host data is fed into the OUT FIFO. The host will send the same amount of data on every frame, i.e. 48 kHz of data based on the host clock. The codec sampling clock is asynchronous to the host clock. This will cause the amount of data in the OUT FIFO to vary. If the amount of data in the FIFO exceeds the high water mark, then the sampling clock is increased. If the data is between the high and low water mark, the sampling clock does not change. If the data is below the low water mark, the sampling clock is decreased.

7.3.1.3 Synchronous Operation

For synchronous operation, the internal clock must be synchronized with the host SOF. The Frame SOF is nominally 1mS. Since there is significant jitter in the SOFs, there is circuitry provided to measure the SOFs over a long period of time to get a more accurate reading. The calculated host frequency is used to calculate the codec sampling clock.

USB7206C

8.0 FUNCTIONAL DESCRIPTIONS

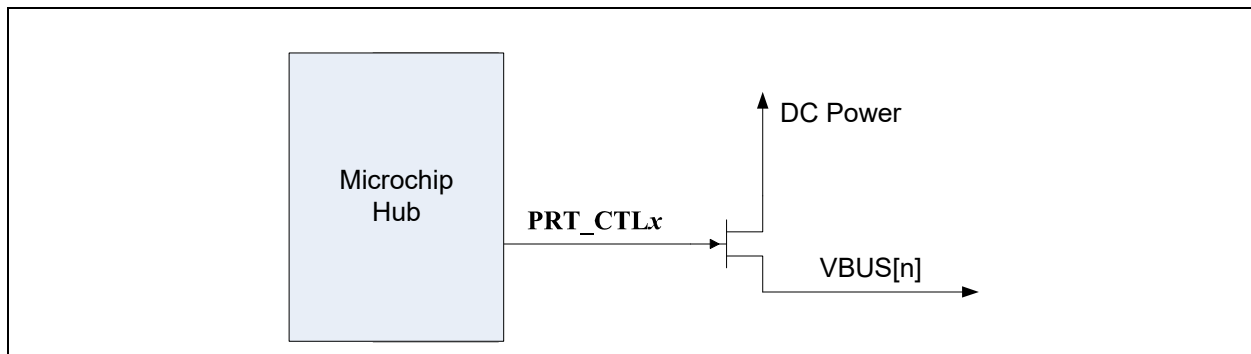
This section details various USB7206C functions, including:

- [Downstream Battery Charging](#)
- [Port Power Control](#)
- [PortSplit](#)
- [FlexConnect](#)
- [USB to GPIO Bridging](#)
- [USB to I2C Bridging](#)
- [USB to SPI Bridging](#)
- [Link Power Management \(LPM\)](#)
- [Resets](#)

8.1 Downstream Battery Charging

The device can be configured by an OEM to have any of the downstream ports support battery charging. The hub's role in battery charging is to provide acknowledgment to a device's query as to whether the hub system supports USB battery charging. The hub silicon does not provide any current or power FETs or any additional circuitry to actually charge the device. Those components must be provided externally by the OEM.

FIGURE 8-1: BATTERY CHARGING EXTERNAL POWER SUPPLY



If the OEM provides an external supply capable of supplying current per the battery charging specification, the hub can be configured to indicate the presence of such a supply from the device. This indication, via the **PRT_CTLx** pins, is on a per port basis. For example, the OEM can configure two ports to support battery charging through high current power FETs and leave the other two ports as standard USB ports.

The port control signals are assigned to programmable pins (**PFx**) and therefore the device must be programmed into specific configurations to enable the signals. Refer to [Section 3.3.4, PF\[31:3\] Configuration \(CFG_STRAP\[2:1\]\)](#) for additional information.

For detailed information on utilizing the battery charging feature, refer to the application note "[USB Battery Charging with Microchip USB720x and USB725x Hubs](#)", which can be found on the Microchip USB7206C product page www.microchip.com/USB7206C.

8.2 Port Power Control

Port power and over-current sense share the same pin (**PRT_CTLx**) for each port. These functions can be controlled directly from the USB hub, or via the processor.

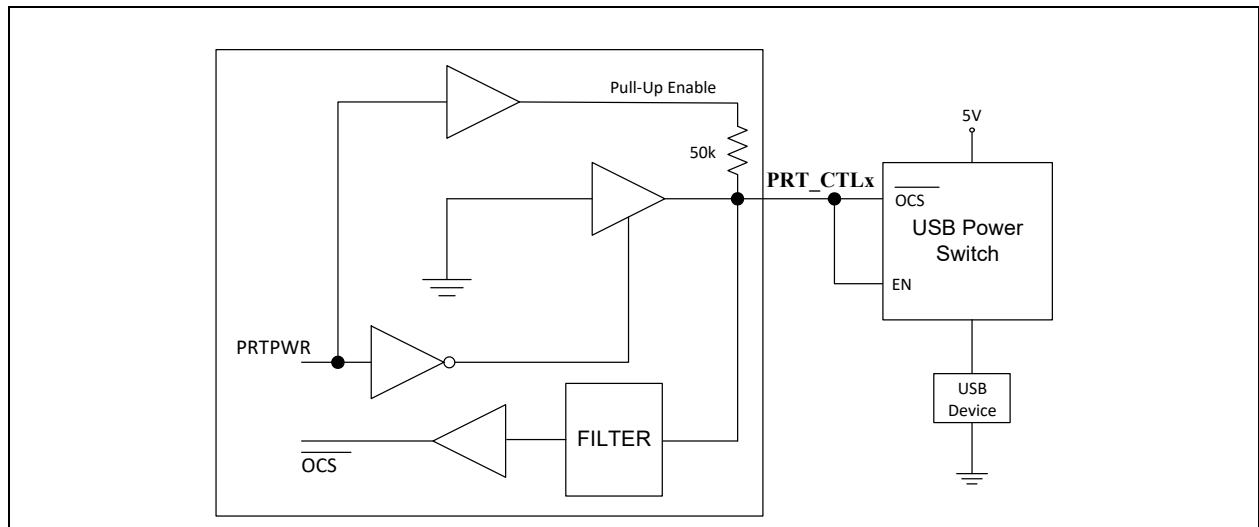
Note: The **PRT_CTLx** function is assigned to programmable function pins (**PFx**) via configuration straps. Refer to [Section 3.3.4, PF\[31:3\] Configuration \(CFG_STRAP\[2:1\]\)](#) for additional information.

8.2.1 PORT POWER CONTROL USING USB POWER SWITCH

When operating in combined mode, the device will have one port power control and over-current sense pin for each downstream port. When disabling port power, the driver will actively drive a '0'. To avoid unnecessary power dissipation, the pull-up resistor will be disabled at that time. When port power is enabled, it will disable the output driver and enable the pull-up resistor, making it an open drain output. If there is an over-current situation, the USB Power Switch will assert the open drain OCS signal. The Schmidt trigger input will recognize that as a low. The open drain output does not interfere. The over-current sense filter handles the transient conditions such as low voltage while the device is powering up.

Note: An external power switch is the required implementation for Type-C ports due to the requirement that VBUS on Type-C ports must be discharged to 0V when no device is attached to the port.

FIGURE 8-2: PORT POWER CONTROL WITH USB POWER SWITCH



8.2.2 PORT POWER CONTROL USING POLY FUSE

When using the device with a poly fuse, there is no need for an output power control. A single port power control and over-current sense for each downstream port is still used from the Hub's perspective. When disabling port power, the driver will actively drive a '0'. This will have no effect as the external diode will isolate pin from the load. When port power is enabled, it will disable the output driver and enable the pull-up resistor. This means that the pull-up resistor is providing 3.3 volts to the anode of the diode. If there is an over-current situation, the poly fuse will open. This will cause the cathode of the diode to go to 0 volts. The anode of the diode will be at 0.7 volts, and the Schmidt trigger input will register this as a low resulting in an over-current detection. The open drain output does not interfere.

Note: Type-C ports may not utilize a Poly-Fuse port power implementation due to the requirements that VBUS on Type-C ports must be discharged to 0V when no device is attached to the port.

The PortSplit feature allows the USB 3.0 and USB 3.2 PHYs associated with a downstream port to be operationally con-

Wavelength (nm)	Excitation (nm)	Fluorescence (nm)	Probe	Concentration (M)	Temperature (°C)	pH	Time (min)	Intensity (a.u.)	Decay Time (ns)	Decay Time (ns)
280	280	340	100	1.0	25	7.4	0	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	10	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	20	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	30	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	40	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	50	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	60	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	70	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	80	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	90	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	100	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	110	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	120	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	130	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	140	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	150	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	160	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	170	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	180	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	190	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	200	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	210	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	220	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	230	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	240	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	250	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	260	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	270	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	280	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	290	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	300	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	310	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	320	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	330	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	340	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	350	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	360	1.0	1.0	1.0
280	280	340	100	1.0	25	7.4	370	1.0	1.0	

Figure 1. The effect of the number of trials on the number of correct responses. The number of correct responses was significantly higher than the number of incorrect responses in all cases. Error bars represent the standard error of the mean.

2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026 2027 2028 2029 2030 2031 2032 2033 2034 2035 2036 2037 2038 2039 2040 2041 2042 2043 2044 2045 2046 2047 2048 2049 2050 2051 2052 2053 2054 2055 2056 2057 2058 2059 2060 2061 2062 2063 2064 2065 2066 2067 2068 2069 2070 2071 2072 2073 2074 2075 2076 2077 2078 2079 2080 2081 2082 2083 2084 2085 2086 2087 2088 2089 2090 2091 2092 2093 2094 2095 2096 2097 2098 2099 2100 2101 2102 2103 2104 2105 2106 2107 2108 2109 2110 2111 2112 2113 2114 2115 2116 2117 2118 2119 2120 2121 2122 2123 2124 2125 2126 2127 2128 2129 2130 2131 2132 2133 2134 2135 2136 2137 2138 2139 2140 2141 2142 2143 2144 2145 2146 2147 2148 2149 2150 2151 2152 2153 2154 2155 2156 2157 2158 2159 2160 2161 2162 2163 2164 2165 2166 2167 2168 2169 2170 2171 2172 2173 2174 2175 2176 2177 2178 2179 2180 2181 2182 2183 2184 2185 2186 2187 2188 2189 2190 2191 2192 2193 2194 2195 2196 2197 2198 2199 2200 2201 2202 2203 2204 2205 2206 2207 2208 2209 2210 2211 2212 2213 2214 2215 2216 2217 2218 2219 2220 2221 2222 2223 2224 2225 2226 2227 2228 2229 2230 2231 2232 2233 2234 2235 2236 2237 2238 2239 2240 2241 2242 2243 2244 2245 2246 2247 2248 2249 2250 2251 2252 2253 2254 2255 2256 2257 2258 2259 2260 2261 2262 2263 2264 2265 2266 2267 2268 2269 2270 2271 2272 2273 2274 2275 2276 2277 2278 2279 2280 2281 2282 2283 2284 2285 2286 2287 2288 2289 2290 2291 2292 2293 2294 2295 2296 2297 2298 2299 2300 2301 2302 2303 2304 2305 2306 2307 2308 2309 2310 2311 2312 2313 2314 2315 2316 2317 2318 2319 2320 2321 2322 2323 2324 2325 2326 2327 2328 2329 2330 2331 2332 2333 2334 2335 2336 2337 2338 2339 2340 2341 2342 2343 2344 2345 2346 2347 2348 2349 2350 2351 2352 2353 2354 2355 2356 2357 2358 2359 2360 2361 2362 2363 2364 2365 2366 2367 2368 2369 2370 2371 2372 2373 2374 2375 2376 2377 2378 2379 2380 2381 2382 2383 2384 2385 2386 2387 2388 2389 2390 2391 2392 2393 2394 2395 2396 2397 2398 2399 2400 2401 2402 2403 2404 2405 2406 2407 2408 2409 2410 2411 2412 2413 2414 2415 2416 2417 2418 2419 2420 2421 2422 2423 2424 2425 2426 2427 2428 2429 2430 2431 2432 2433 2434 2435 2436 2437 2438 2439 2440 2441 2442 2443 2444 2445 2446 2447 2448 2449 2450 2451 2452 2453 2454 2455 2456 2457 2458 2459 2460 2461 2462 2463 2464 2465 2466 2467 2468 2469 2470 2471 2472 2473 2474 2475 2476 2477 2478 2479 2480 2481 2482 2483 2484 2485 2486 2487 2488 2489 2490 2491 2492 2493 2494 2495 2496 2497 2498 2499 2500 2501 2502 2503 2504 2505 2506 2507 2508 2509 2510 2511 2512 2513 2514 2515 2516 2517 2518 2519 2520 2521 2522 2523 2524 2525 2526 2527 2528 2529 2530 2531 2532 2533 2534 2535 2536 2537 2538 2539 2540 2541 2542 2543 2544 2545 2546 2547 2548 2549 2550 2551 2552 2553 2554 2555 2556 2557 2558 2559 2560 2561 2562 2563 2564 2565 2566 2567 2568 2569 2570 2571 2572 2573 2574 2575 2576 2577 2578 2579 2580 2581 2582 2583 2584 2585 2586 2587 2588 2589 2590 2591 2592 2593 2594 2595 2596 2597 2598 2599 2600 2601 2602 2603 2604 2605 2606 2607 2608 2609 2610 2611 2612 2613 2614 2615 2616 2617 2618 2619 2620 2621 2622 2623 2624 2625 2626 2627 2628 2629 2630 2631 2632 2633 2634 2635 2636 2637 2638 2639 2640 2641 2642 2643 2644 2645 2646 2647 2648 2649 2650 2651 2652 2653 2654 2655 2656 2657 2658 2659 2660 2661 2662 2663 2664 2665 2666 2667 2668 2669 2670 2671 2672 2673 2674 2675 2676 2677 2678 2679 2680 2681 2682 2683 2684 2685 2686 2687 2688 2689 2690 2691 2692 2693 2694 2695 2696 2697 2698 2699 2700 2701 2702 2703 2704 2705 2706 2707 2708 2709 2710 2711 2712 2713 2714 2715 2716 2717 2718 2719 2720 2721 2722 2723 2724 2725 2726 2727 2728 2729 2730 2731 2732 2733 2734 2735 2736 2737 2738 2739 2740 2741 2742 2743 2744 2745 2746 2747 2748 2749 2750 2751 2752 2753 2754 2755 2756 2757 2758 2759 2760 2761 2762 2763 2764 2765 2766 2767 2768 2769 2770 2771 2772 2773 2774 2775 2776 2777 2778 2779 2780 2781 2782 2783 2784 2785 2786 2787 2788 2789 2790 2791 2792 2793 2794 2795 2796 2797 2798 2799 2800 2801 2802 2803 2804 2805 2806 2807 2808 2809 2810 2811 2812 2813 2814 2815 2816 2817 2818

8.5 USB to GPIO Bridging

The USB to GPIO bridging feature provides system designers expanded system control and potential BOM reduction. General Purpose Input/Outputs (GPIOs) may be used for any general 3.3V level digital control and input functions.

Commands may be sent from the USB Host to the internal Hub Feature Controller device in the Microchip hub to perform the following functions:

- Set the direction of the GPIO (input or output)
- Enable a pull-up resistor
- Enable a pull-down resistor
- Read the state
- Set the state

For detailed information on utilizing the USB to GPIO bridging feature, refer to the application note “*USB to GPIO Bridging with Microchip USB720x and USB725x Hubs*”, which can be found on the Microchip USB7206C product page at www.microchip.com/USB7206C.

8.6 USB to I²C Bridging

The USB to I²C bridging feature provides system designers expanded system control and potential BOM reduction. The use of a separate USB to I²C device is no longer required and a downstream USB port is not lost, as occurs when a standalone USB to I²C device is implemented.

Commands may be sent from the USB Host to the internal Hub Feature Controller device in the Microchip hub to perform the following functions:

- Configure I²C Pass-Through Interface
- I²C Write
- I²C Read

For detailed information on utilizing the USB to I²C bridging feature, refer to the application note “*USB to I²C Bridging with Microchip USB720x and USB725x Hubs*”, which can be found on the Microchip USB7206C product page at www.microchip.com/USB7206C.

8.7 USB to SPI Bridging

The USB to SPI bridging feature provides system designers expanded system control and potential BOM reduction. The use of a separate USB to SPI device is no longer required and a downstream USB port is not lost, as occurs when a standalone USB to SPI device is implemented.

Commands may be sent from the USB Host to the internal Hub Feature Controller device in the Microchip hub to perform the following functions:

- Enable SPI Pass-Through Interface
- SPI Write/Read
- Disable SPI Pass-Through Interface

For detailed information on utilizing the USB to SPI bridging feature, refer to the application note “*USB to SPI Bridging with Microchip USB720x and USB725x Hubs*”, which can be found on the Microchip USB7206C product page at www.microchip.com/USB7206C.

8.8 Link Power Management (LPM)

The device supports the L0 (On), L1 (Sleep), and L2 (Suspend) link power management states. These supported LPM states offer low transitional latencies in the tens of microseconds versus the much longer latencies of the traditional USB suspend/resume in the tens of milliseconds. The supported LPM states are detailed in [Table 8-1](#).

TABLE 8-1: LPM STATE DEFINITIONS

State	Description	Entry/Exit Time to L0
L2	Suspend	Entry: ~3 ms Exit: ~2 ms (from start of RESUME)
L1	Sleep	Entry: <10 μ s Exit: <50 μ s
L0	Fully Enabled (On)	—

8.9 Resets

The device includes the following chip-level reset sources:

- [Power-On Reset \(POR\)](#)
- [External Chip Reset \(RESET_N\)](#)
- [USB Bus Reset](#)

8.9.1 POWER-ON RESET (POR)

A power-on reset occurs whenever power is initially supplied to the device, or if power is removed and reapplied to the device. A timer within the device will assert the internal reset per the specifications listed in [Section 9.6.2, Power-On and Configuration Strap Timing](#).

8.9.2 EXTERNAL CHIP RESET (RESET_N)

A valid hardware reset is defined as assertion of **RESET_N**, after all power supplies are within operating range, per the specifications in [Section 9.6.3, Reset and Configuration Strap Timing](#). While reset is asserted, the device (and its associated external circuitry) enters Standby Mode and consumes minimal current.

Assertion of **RESET_N** causes the following:

1. The PHY is disabled and the differential pairs will be in a high-impedance state.
2. All transactions immediately terminate; no states are saved.
3. All internal registers return to the default state.
4. The external crystal oscillator is halted.
5. The PLL is halted.

Note: All power supplies must have reached the operating levels mandated in [Section 9.2, Operating Conditions**](#), prior to (or coincident with) the assertion of **RESET_N**.

8.9.3 USB BUS RESET

In response to the upstream port signaling a reset to the device, the device performs the following:

1. Sets default address to 0.
2. Sets configuration to Unconfigured.
3. Moves device from suspended to active (if suspended).
4. Complies with the USB Specification for behavior after completion of a reset sequence.

The host then configures the device in accordance with the USB Specification.

Note: The device does not propagate the upstream USB reset to downstream devices.

9.0 OPERATIONAL CHARACTERISTICS

9.1 Absolute Maximum Ratings*

Digital Core Supply Voltage (V _{CORE}) (Note 1)	-0.5V to +1.25V
+3.3V Supply Voltage (V _{DD33}) (Note 1)	-0.5V to +4.6V
Positive voltage on input signal pins, with respect to ground (Note 2)	+4.6V
Negative voltage on input signal pins, with respect to ground	-0.5V
Positive voltage on XTALI/CLK_IN, with respect to ground	+3.63V
Positive voltage on USB DP/DM signal pins, with respect to ground	+6.0V
Positive voltage on USB 3.2 Gen 2 USB3UP_XXXX and USB3DN_XXXX signal pins, with respect to ground	1.25V
Storage Temperature	-55°C to +150°C
Junction Temperature	+125°C
Lead Temperature Range	Refer to JEDEC Spec. J-STD-020
HBM ESD Performance	+/-3.5 kV

Note 1: When powering this device from laboratory or system power supplies, it is important that the absolute maximum ratings not be exceeded or device failure can result. Some power supplies exhibit voltage spikes on their outputs when AC power is switched on or off. In addition, voltage transients on the AC power line may appear on the DC output. If this possibility exists, it is suggested that a clamp circuit be used.

Note 2: This rating does not apply to the following pins: All USB DM/DP pins, XTALI/CLK_IN, and XTALO.

*Stresses exceeding those listed in this section could cause permanent damage to the device. This is a stress rating only. Exposure to absolute maximum rating conditions for extended periods may affect device reliability. Functional operation of the device at any condition exceeding those indicated in [Section 9.2, Operating Conditions**](#), [Section 9.5, DC Specifications](#), or any other applicable section of this specification is not implied.

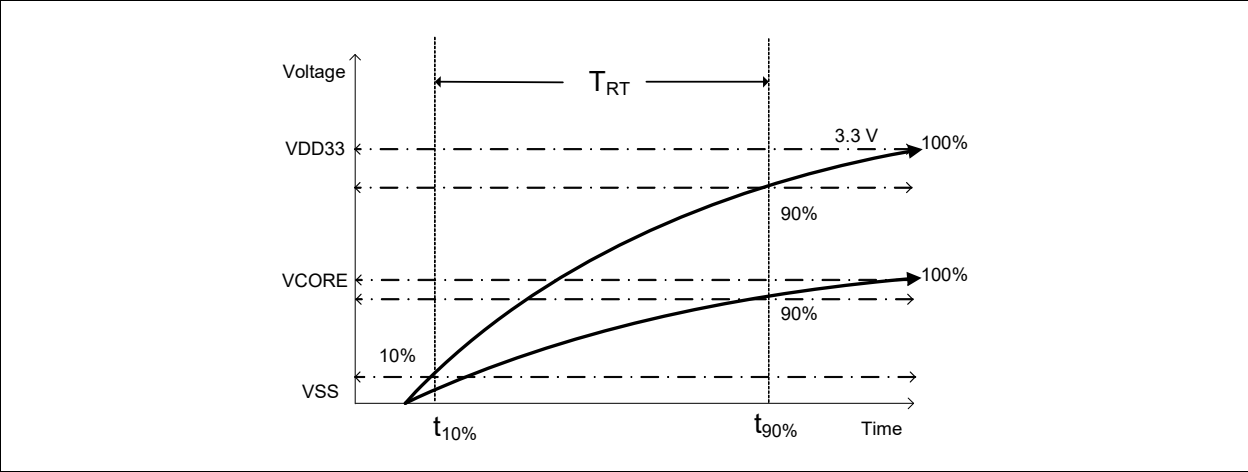
9.2 Operating Conditions**

Digital Core Supply Voltage (V _{CORE})	+1.09V to +1.21V
+3.3V Supply Voltage (V _{DD33})	+3.0V to +3.6V
Input Signal Pins Voltage (Note 2)	-0.3V to +3.6V
XTALI/CLK_IN Voltage	-0.3V to +3.6V
USB 2.0 DP/DM Signal Pins Voltage	-0.3V to +5.5V
USB 3.2 Gen 2 USB3UP_XXXX and USB3DN_XXXX Signal Pins Voltage	-0.3V to +1.21V
Ambient Operating Temperature in Still Air (T _A)	Note 3
Digital Core Supply Voltage Rise Time (T _{RT} in Figure 9-1)	5 ms
+3.3V Supply Voltage Rise Time (T _{RT} in Figure 9-1)	5 ms

Note 3: 0°C to +70°C for commercial version, -40°C to +85°C for industrial version.

**Proper operation of the device is guaranteed only within the ranges specified in this section. Do not drive input signals without power supplied to the device.

FIGURE 9-1: SUPPLY RISE TIME MODEL



Note: The Power Supply Rise time requirement does not apply if the **RESET_N** signal is held low during power on and released after power levels rise and stabilize above the power on thresholds, or if the **RESET_N** signal is toggled after power supplies become stable.

9.3 Package Thermal Specifications

TABLE 9-1: PACKAGE THERMAL PARAMETERS

Symbol	°C/W	Velocity (Meters/s)
Θ_{JA}	19	0
	16	1
	14	2.5
Ψ_{JT}	0.1	0
	0.1	1
Ψ_{JB}	9	0
Θ_{JC}	1.3	0
	1.3	1
Θ_{JB}	10	—

Note: Thermal parameters are measured or estimated for devices in a multi-layer 2S2P PCB per JESDN51. For industrial applications, the USB7206C requires multi-layer 2S4P PCB power dissipation.

9.4 Power Consumption

This section details the power consumption of the device as measured during various modes of operation. Power dissipation is determined by temperature, supply voltage, and external source/sink requirements.

TABLE 9-2: DEVICE POWER CONSUMPTION

	Typical (mA) @ 25°C		Typical Power (mW)
	VCORE (1.15V)	VDD33 (3.3V)	
Global Suspend	5.4	2.3	14
VBUS Off	5.3	6.4	27
Reset	4.2	—	5
Data for Calculating Active Transfer Current			
Upstream Port Link Speed Base Currents			
SS+ Current	410	30.8	—
SS Current	370	27.3	—
HS Current	58	19.7	—
Additional Current Per Enabled Port			
SS+ Current	179	11.1	—
SS Current	143	9.1	—
HS Current	1	10.8	—
Example Active Data Transfer Current Calculation: 1 SS+ Port and 2 HS Ports			
Active Data Transfer Current (mA @ 3.3V)	$\{30.8\} + \{1 * 11.1\} + \{2 * 10.8\} = 63.5$		
Active Data Transfer Current (mA @ 1.15V)	$\{410\} + \{1 * 179\} + \{2 * 1\} = 591$		

Note: In the Active Idle and Active Data Transfer sections of [Table 9-2](#), the various port configurations are indicated via the following acronyms:

SS+ = USB 3.2 SuperSpeed+ (Gen 2)

SS = USB 3.2 SuperSpeed (Gen 1)

HS = USB 2.0 High Speed

USB7206C

9.5 DC Specifications

TABLE 9-3: I/O DC ELECTRICAL CHARACTERISTICS

Parameter	Symbol	Min	Typical	Max	Units	Notes
I Type Input Buffer						
Low Input Level	V _{IL}	—	—	0.9	V	—
High Input Level	V _{IH}	2.1	—	—	V	
IS Type Input Buffer						
Low Input Level	V _{IL}	—	—	0.9	V	—
High Input Level	V _{IH}	2.1	—	—	V	
Schmitt Trigger Hysteresis (V _{IHT} - V _{ILT})	V _{HYS}	100	160	240	mV	
O12 Type Output Buffer						
Low Output Level	V _{OL}	—	—	0.4	V	I _{OL} = 12 mA
High Output Level	V _{OH}	VDD33-0.4	—	—	V	I _{OH} = -12 mA
OD12 Type Output Buffer						
Low Output Level	V _{OL}	—	—	0.4	V	I _{OL} = 12 mA
ICLK Type Input Buffer (XTALI Input)						
Low Input Level	V _{IL}	—	—	0.35	V	Note 4
High Input Level	V _{IH}	1.1	—	—	V	
IO-U Type Buffer (See Note 5)	—	—	—	—	—	Note 5

Note 4: XTALI can optionally be driven from a 25 MHz singled-ended clock oscillator.

Note 5: Refer to the USB 3.2 Gen 2 Specification for USB DC electrical characteristics.

9.6 AC Specifications

This section details the various AC timing specifications of the device.

9.6.1 POWER SUPPLY AND RESET_N SEQUENCE TIMING

There is no specific requirement for power sequencing of VDD33 and V_{CORE} for device operation. Figure 9-2 illustrates the recommended power supply sequencing for ensuring long term reliability of the device. V_{CORE} should rise after or at the same time as VDD33. Similarly, RESET_N and/or VBUS_DET should rise after or at the same time as VDD33. VBUS_DET and RESET_N do not have any other timing dependencies. The rise times for V_{CORE} and VDD33 are provided in Section 9.2, Operating Conditions** and Figure 9-1.

FIGURE 9-2: POWER SUPPLY AND RESET_N SEQUENCE TIMING

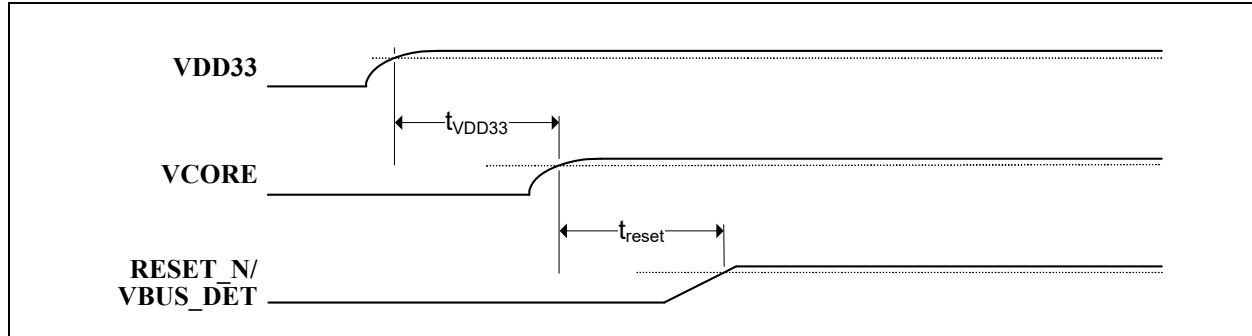


TABLE 9-4: POWER SUPPLY AND RESET_N SEQUENCE TIMING

Symbol	Description	Min	Typ	Max	Units
t_{VDD33}	VDD33 to V _{CORE} rise delay	0	—	—	ms
t_{reset}	VDD33 to RESET_N/VBUS_DET rise delay	0	—	—	ms

9.6.2 POWER-ON AND CONFIGURATION STRAP TIMING

Figure 9-3 illustrates the configuration strap valid timing requirements in relation to power-on, for applications where RESET_N is not used at power-on. In order for valid configuration strap values to be read at power-on, the following timing requirements must be met. The operational levels (V_{opp}) for the external power supplies are detailed in Section 9.2, Operating Conditions**.

FIGURE 9-3: POWER-ON CONFIGURATION STRAP VALID TIMING

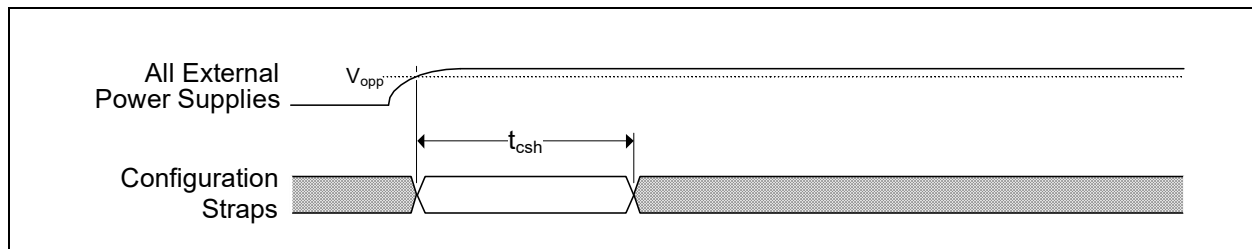


TABLE 9-5: POWER-ON CONFIGURATION STRAP LATCHING TIMING

Symbol	Description	Min	Typ	Max	Units
t_{csh}	Configuration strap hold after external power supplies at operational levels	1	—	—	ms

Device configuration straps are also latched as a result of RESET_N assertion. Refer to Section 9.6.3, Reset and Configuration Strap Timing for additional details.

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9.6.3 RESET AND CONFIGURATION STRAP TIMING

Figure 9-4 illustrates the RESET_N pin timing requirements and its relation to the configuration strap pins. Assertion of RESET_N is not a requirement. However, if used, it must be asserted for the minimum period specified. Refer to Section 8.9, Resets for additional information on resets. Refer to Section 3.3, Configuration Straps and Programmable Functions for additional information on configuration straps.

FIGURE 9-4: RESET_N CONFIGURATION STRAP TIMING

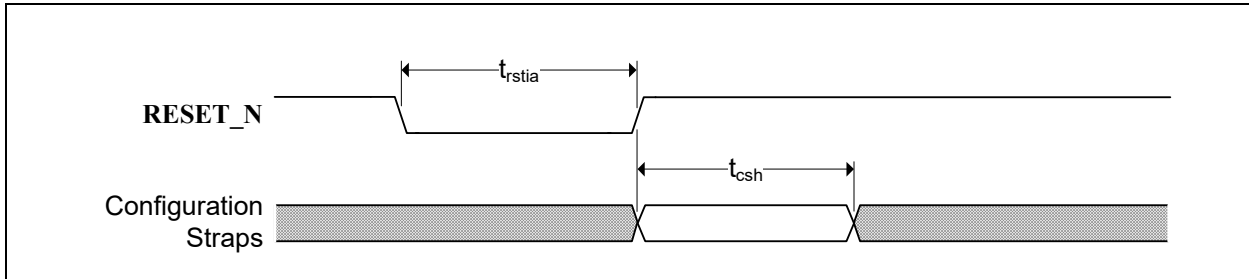


TABLE 9-6: RESET_N CONFIGURATION STRAP TIMING

Symbol	Description	Min	Typ	Max	Units
t_{rstia}	RESET_N input assertion time	5	—	—	μs
t_{csh}	Configuration strap pins hold after RESET_N deassertion	1	—	—	ms

Note: The clock input must be stable prior to RESET_N deassertion.

Configuration strap latching and output drive timings shown assume that the Power-On reset has finished first otherwise the timings in Section 9.6.2, Power-On and Configuration Strap Timing apply.

9.6.4 POWER-ON OR RESET TO SMBUS SLAVE READY TIMING

Figure 9-5 illustrates the SMBus Slave interface readiness in relation to power-on or de-assertion of RESET_N. In order to ensure reliable SMBus slave operation, the SMBus master must allow the bus to remain idle until t_{SMBUS_RDY} timing has been met. The operational levels (V_{opp}) for the external power supplies are detailed in Section 9.2, Operating Conditions**.

FIGURE 9-5: POWER-ON OR RESET TO SMBUS SLAVE READY TIMING

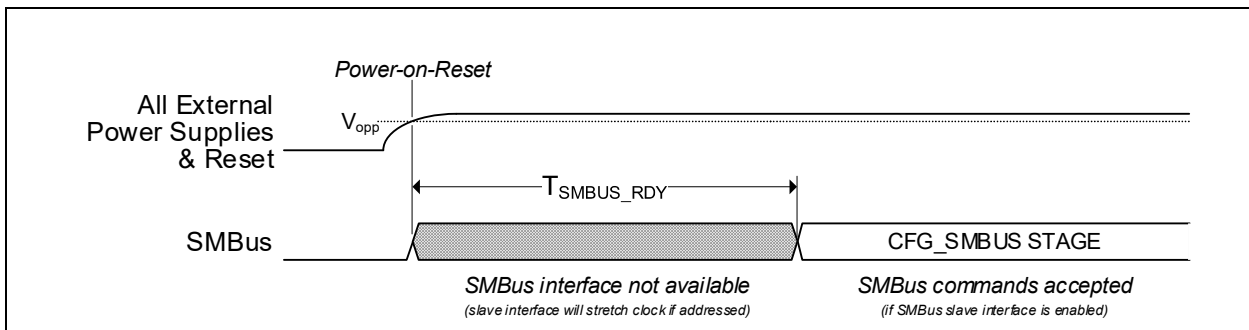


TABLE 9-7: POWER-ON OR RESET TO SMBUS SLAVE READY TIMING

Symbol	Description	Min	Typ	Max	Units
t_{SMBUS_RDY}	Power-on or RESET_N deassertion to SMBus ready	40	—	—	ms

9.6.5 USB ATTACH COMMAND TO SMBUS SLAVE READY TIMING

Figure 9-6 illustrates the SMBus Slave interface readiness in relation to ACK of the Slave interface to the “USB Attach with SMBus Runtime Access” (AA56h) from the SMBus Master. In order to ensure reliable SMBus slave operation, the SMBus master must allow the bus to remain idle after issuing the “USB Attach with SMBus Runtime Access” until t_{ATTACH_RDY} timing has been met.

Note: When accessing SMBus during runtime, it is critical to force some clocks to stay on. If this step is not taken, the SMBus slave interface will not be accessible while the hub is placed into a Suspend state by the host.

FIGURE 9-6: USB ATTACH COMMAND TO SMBUS SLAVE READY TIMING

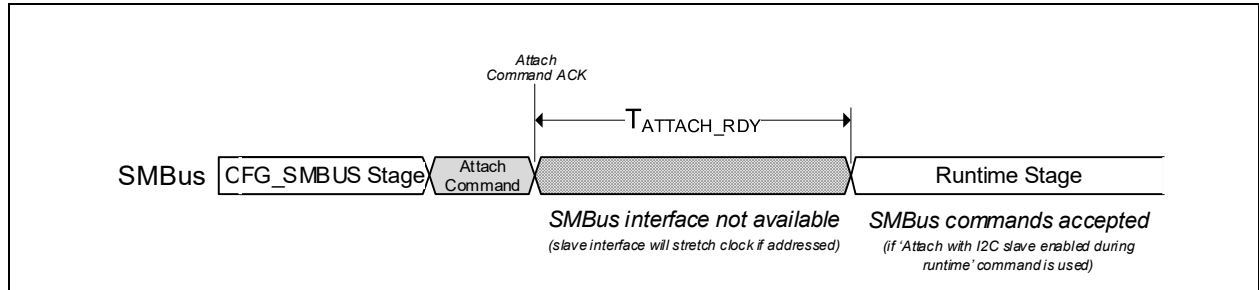


TABLE 9-8: USB ATTACH COMMAND TO SMBUS SLAVE READY TIMING

Symbol	Description	Min	Typ	Max	Units
t_{ATTACH_RDY}	USB Attach command to SMBus ready (Note 6)	11.5	—	—	ms

Note 6: The t_{ATTACH_RDY} values are preliminary and subject to change.

9.6.6 USB TIMING

All device USB signals conform to the voltage, power, and timing characteristics/specifications as set forth in the *Universal Serial Bus Specification*. Please refer to the *Universal Serial Bus Revision 3.2 Specification*, available at <http://www.usb.org/developers/docs>.

9.6.7 SMBUS TIMING

All device SMBus signals conform to the voltage, power, and timing characteristics/specifications as set forth in the *System Management Bus Specification*. Please refer to the *System Management Bus Specification*, Version 1.0, available at <http://smbus.org/specs>.

9.6.8 I²C TIMING

All device I²C signals conform to the 100 kHz Standard-mode (Sm) and 400 kHz Fast Mode (Fm) voltage, power, and timing characteristics/specifications as set forth in the *I²C-Bus Specification*. Please refer to the *I²C-Bus Specification*, available at http://www.nxp.com/documents/user_manual/UM10204.pdf.

9.6.9 I²S TIMING

All device I²S signals conform to the voltage, power, and timing characteristics/specifications as set forth in the *I²S-Bus Specification*. Please refer to the *I²S-Bus Specification*, available at www.sparkfun.com/datasheets/BreakoutBoards/I2SBUS.pdf.

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9.6.10 SPI/SQI MASTER TIMING

This section specifies the SPI/SQI master timing requirements for the device.

FIGURE 9-7: SPI/SQI MASTER TIMING

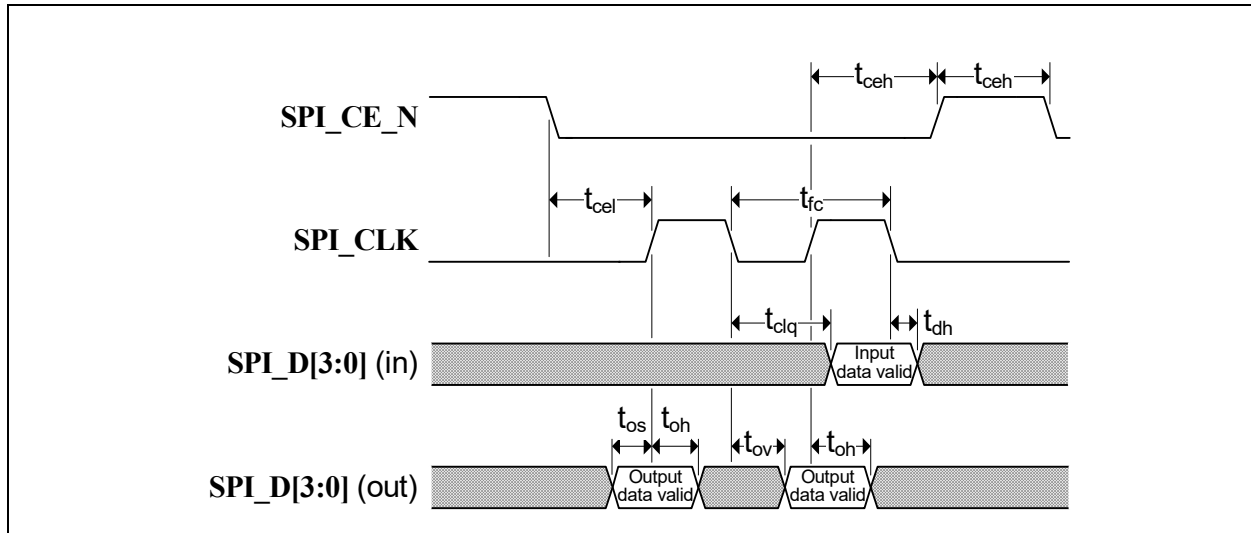


TABLE 9-9: SPI/SQI MASTER TIMING (30 MHZ OPERATION)

Symbol	Description	Min	Typ	Max	Units
t_{fc}	Clock frequency	—	—	30	MHz
t_{ceh}	Chip enable (SPI_CE_N) high time	100	—	—	ns
t_{clq}	Clock to input data	—	—	13	ns
t_{dh}	Input data hold time	0	—	—	ns
t_{os}	Output setup time	5	—	—	ns
t_{oh}	Output hold time	5	—	—	ns
t_{ov}	Clock to output valid	4	—	—	ns
t_{cel}	Chip enable (SPI_CE_N) low to first clock	12	—	—	ns
t_{ceh}	Last clock to chip enable (SPI_CE_N) high	12	—	—	ns

TABLE 9-10: SPI/SQI MASTER TIMING (60 MHZ OPERATION)

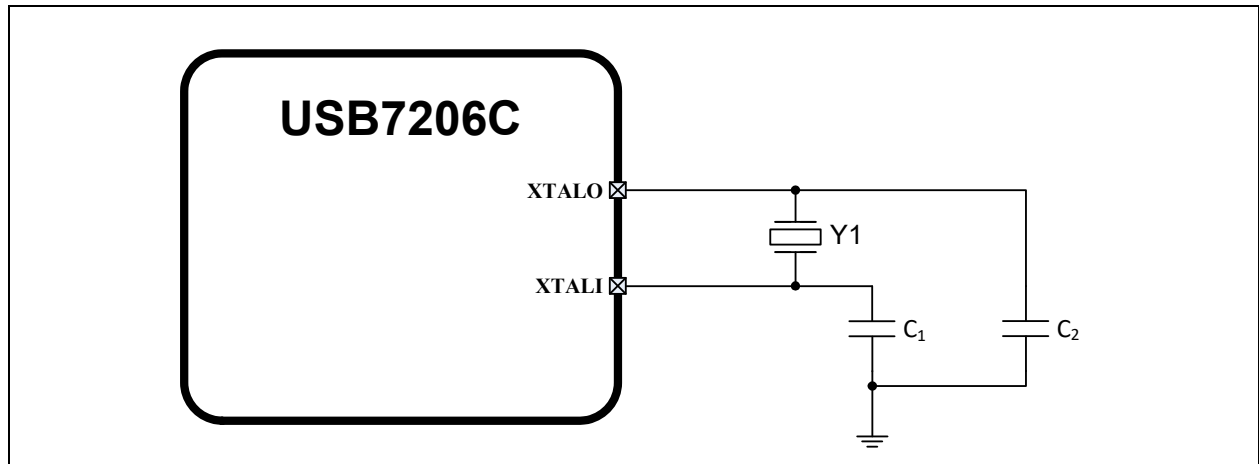
Symbol	Description	Min	Typ	Max	Units
t_{fc}	Clock frequency	—	—	60	MHz
t_{ceh}	Chip enable (SPI_CE_N) high time	50	—	—	ns
t_{clq}	Clock to input data	—	—	9	ns
t_{dh}	Input data hold time	0	—	—	ns
t_{os}	Output setup time	5	—	—	ns
t_{oh}	Output hold time	5	—	—	ns
t_{ov}	Clock to output valid	4	—	—	ns
t_{cel}	Chip enable (SPI_CE_N) low to first clock	12	—	—	ns
t_{ceh}	Last clock to chip enable (SPI_CE_N) high	12	—	—	ns

9.7 Clock Specifications

The device can accept either a 25 MHz crystal or a 25 MHz single-ended clock oscillator input. If the single-ended clock oscillator method is implemented, XTALO should be left unconnected and XTALI/CLK_IN should be driven with a nominal 0-3.3V clock signal. The input clock duty cycle is 40% minimum, 50% typical and 60% maximum.

It is recommended that a crystal utilizing matching parallel load capacitors be used for the crystal input/output signals (XTALI/XTALO). The following circuit design (Figure 9-8) and specifications (Table 9-11) are required to ensure proper operation.

FIGURE 9-8: 25 MHZ CRYSTAL CIRCUIT



9.7.1 CRYSTAL SPECIFICATIONS

It is recommended that a crystal utilizing matching parallel load capacitors be used for the crystal input/output signals (XTALI/XTALO). Refer to Table 9-11 for the recommended crystal specifications.

TABLE 9-11: CRYSTAL SPECIFICATIONS

Parameter	Symbol	Min	Nom	Max	Units	Notes
Crystal Cut	AT, typ					—
Crystal Oscillation Mode	Fundamental Mode					—
Crystal Calibration Mode	Parallel Resonant Mode					—
Frequency	F_{fund}	—	25.000	—	MHz	—
Frequency Tolerance @ 25°C	F_{tol}	—	—	±50	PPM	—
Frequency Stability Over Temp	F_{temp}	—	—	±50	PPM	—
Frequency Deviation Over Time	F_{age}	—	±3 to 5	—	PPM	Note 7
Total Allowable PPM Budget		—	—	±100	PPM	—
Shunt Capacitance	C_O	—	7 typ	—	pF	—
Load Capacitance	C_L	—	20 typ	—	pF	—
Drive Level	P_W	100	—	—	μW	—
Equivalent Series Resistance	R_1	—	—	60	Ω	—
Operating Temperature Range	—	Note 8	—	Note 9	°C	—
XTALI/CLK_IN Pin Capacitance	—	—	3 typ	—	pF	Note 10
XTALO Pin Capacitance	—	—	3 typ	—	pF	Note 10

Note 7: Frequency Deviation Over Time is also referred to as Aging.

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Note 8: 0 °C for commercial version, -40 °C for industrial version.

Note 9: +70 °C for commercial version, +85 °C for industrial version.

Note 10: This number includes the pad, the bond wire and the lead frame. PCB capacitance is not included in this value. The XTALI/CLK_IN pin, XTALO pin and PCB capacitance values are required to accurately calculate the value of the two external load capacitors. These two external load capacitors determine the accuracy of the 25.000 MHz frequency.

9.7.2 EXTERNAL REFERENCE CLOCK (CLK_IN)

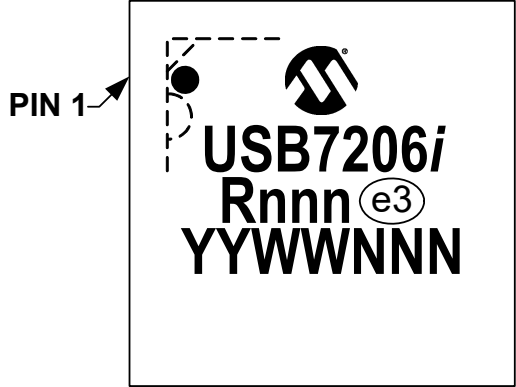
When using an external reference clock, the following clock characteristics are required:

- 25 MHz
- 50% duty cycle $\pm 10\%$, ± 100 ppm
- Jitter < 100 ps RMS

10.0 PACKAGE OUTLINE

10.1 Package Marking Information

100-VQFN (12x12 mm)



Legend:	<i>i</i>	Temperature range designator (Blank = commercial, <i>i</i> = industrial)
	R	Product revision
	nnn	Internal code
	e3	Pb-free JEDEC® designator for Matte Tin (Sn)
	YY	Year code (last two digits of calendar year)
	WW	Week code (week of January 1 is week '01')
	NNN	Alphanumeric traceability code
Note: In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line, thus limiting the number of available characters for customer-specific information.		

* Standard device marking consists of Microchip part number, year code, week code and traceability code. For device marking beyond this, certain price adders apply. Please check with your Microchip Sales Office. For QTP devices, any special marking adders are included in QTP price.

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10.2 100-VQFN Package Drawings

Note: For the most current package drawings, see the Microchip Packaging Specification at: <http://www.microchip.com/packaging>.

FIGURE 10-1: 100-VQFN PACKAGE (DRAWING)

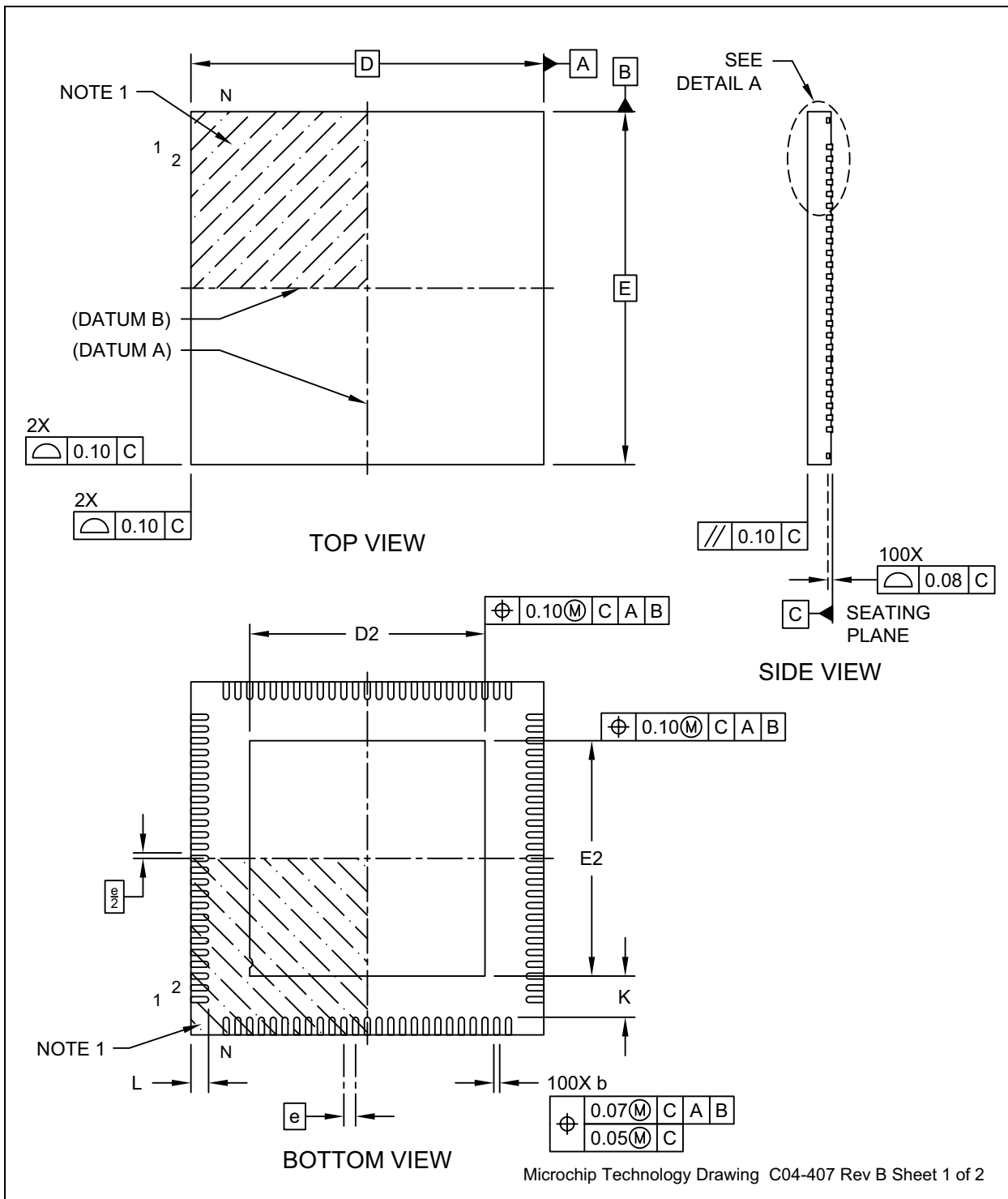
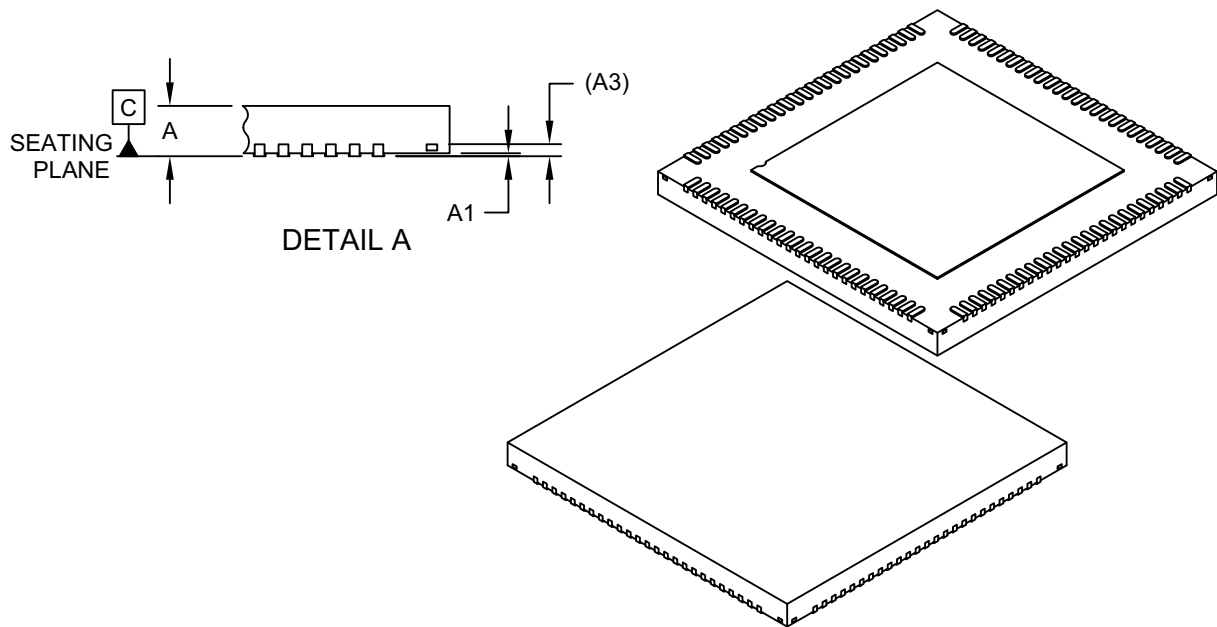


FIGURE 10-2: 100-VQFN PACKAGE (DIMENSIONS)



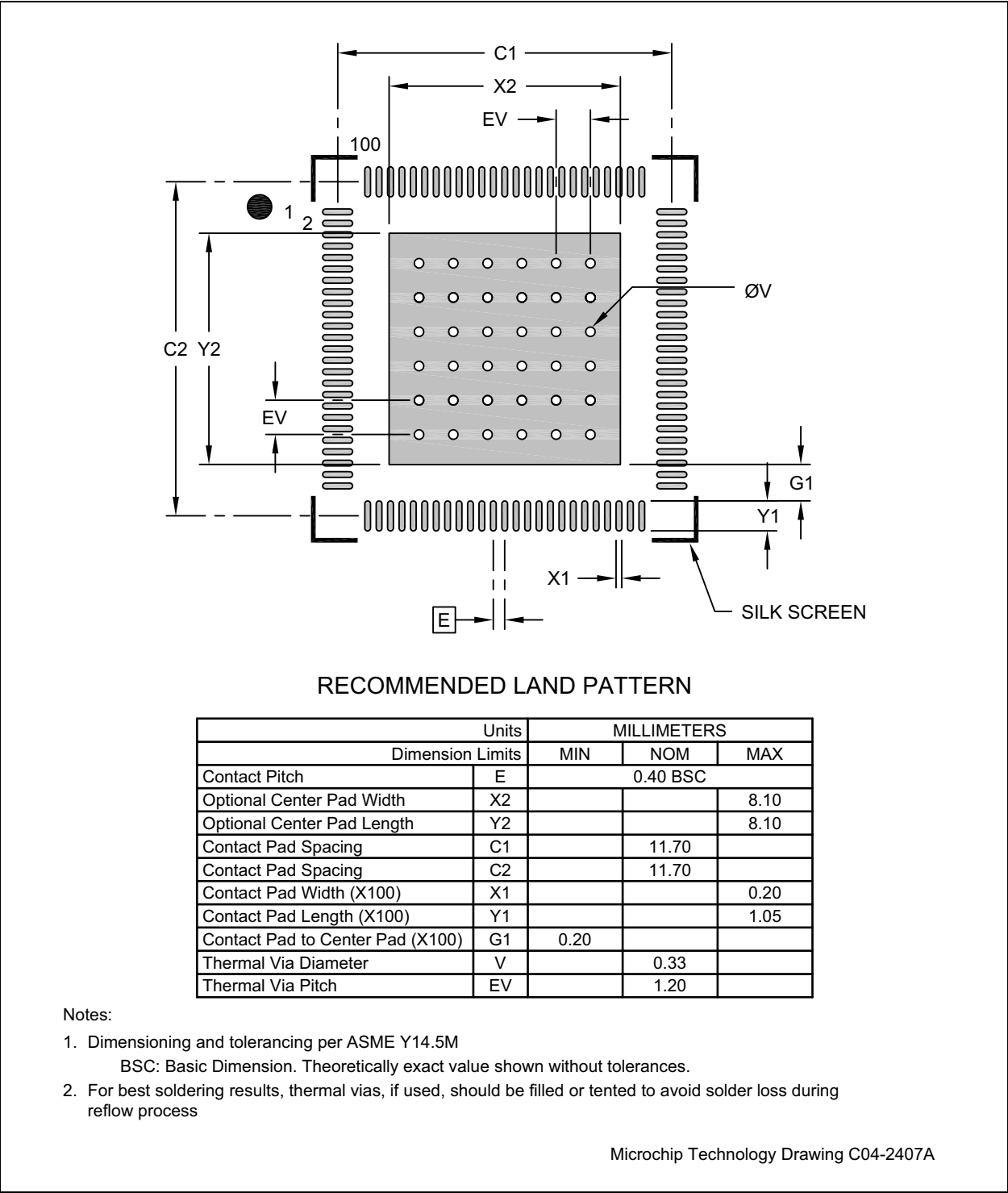
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	100		
Pitch	e	0.40 BSC		
Overall Height	A	0.80	0.85	0.90
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.203 REF		
Overall Length	D	12.00 BSC		
Exposed Pad Length	D2	7.90	8.00	8.10
Overall Width	E	12.00 BSC		
Exposed Pad Width	E2	7.90	8.00	8.10
Terminal Width	b	0.15	0.20	0.25
Terminal Length	L	0.50	0.60	0.70
Terminal-to-Exposed-Pad	K	1.30	-	-

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

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FIGURE 10-3: 100-VQFN PACKAGE (LAND PATTERN)



APPENDIX A: REVISION HISTORY

TABLE A-1: REVISION HISTORY

Revision Level & Date	Section/Figure/Entry	Correction
00003850E (03-06-26)	Section 5.1.8, Hub Connect Stage (USB_ATTACH)	Section rewritten for clarity. Previously stated: "Once the hub registers are updated through default values, SMBus master, and OTP, the device firmware will enable attaching the USB host by setting the USB_ATTACH bit in the HUB_CMD_STAT register (for USB 2.0) and the USB3_HUB_ENABLE bit (for USB 3.2). The device will remain in the Hub Connect stage indefinitely." Now states: "After updating the hub registers in CFG_OTP stage, the firmware enables the USB host attachment by setting the USB_ATTACH bit in the HUB_CMD_STAT register for USB 2.0 and the USB3_HUB_ENABLE bit for USB 3.2. The device remains in this USB_ATTACH stage indefinitely until it gets enumerated by a USB Host."
	Cover	Added trademark symbol to PHYBoost™ and VariSense™
00003850D (01-15-25)	Highlights	Added Celsius temperature ranges to description of Commercial and Industrial support.
	Key Benefits	Added Windows 11 compatibility.
	Table 3-1, "Pin Descriptions", Table 3-5, "PF[31:3] Function Assignment", Section 9.6.1, Power Supply and RESET_N Sequence Timing, Figure 9-4, "RESET_N Configuration Strap Timing", Table 9-6, "RESET_N Configuration Strap Timing"	VBUS_MON_UP pin replaced/renamed with pin VBUS_DET.
00003850C (05-22-24)	Section 4.0, Device Connections, Section 5.0, Modes of Operation, Section 6.0, Device Configuration, Section 7.0, Device Interfaces	Reinserted chapters not present in previous revision.
	Highlights	FlexConnect feature bullet changed from "The roles of the upstream and all downstream ports are reversible on command" to "The roles of the upstream and downstream ports are reversible on command".

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TABLE A-1: REVISION HISTORY (CONTINUED)

Revision Level & Date	Section/Figure/Entry	Correction
	Section 5.1.7, OTP Configuration Stage (CFG_OTP)	Section rewritten for clarity. Previously stated: "Once the SOC has indicated that it is done with configuration, all configuration data is combined in this stage. The default data, the SOC configuration data, and the OTP data are all combined in the firmware and the device is programmed." Now states: "All configuration data from CFG_ROM, CFG_STRAP, CFG_SMBUS stages, and OTP or Pseudo OTP (when SPI firmware is used) are combined in this stage. If a register is modified in both the CFG_SMBUS stage and OTP/Pseudo OTP, the OTP data overwrites SMBus data when executing from ROM firmware, while SMBus data overwrites Pseudo OTP data when executing from SPI firmware. Note that when executing from SPI firmware, the internal OTP content is ignored except for USB2/USB3 UUID data and trim registers for PHY settings."
	Figure 5-1	Updated to remove Configuration 1.
	Section 8.4, FlexConnect	First sentence changed from "The device allows the upstream port to be swapped with any downstream port, enabling any USB port to assume the role of USB host at any time during hub operation" to "The device allows the upstream port to be swapped with a downstream port (all downstream ports except the last), enabling another USB port to assume the role of USB host at any time during hub operation."
	Figure 9-5 and Figure 9-6	Changed "SOC_CFG" to "CFG_SMBUS" in both diagrams.
00003850B (01-08-24)	Table 3-5	Updated Function Assignment for PF29 from "default function is not used" to "GPIO93".
	USB7206C 100-VQFN Pin Assignments	Added Pin description of Pin 71
	Power/Ground	Updated VDD33 pin description from "+3.3V power and internal regulator input" to "+3.3V power supply input".
	Package Marking Information	Updated package top marking from "USB7206C to USB7206" to correctly display silicon revision.
	Section 9.1, Absolute Maximum Ratings*	Updated Digital Core Supply Voltage and Positive voltage on USB 3.2 Gen 2 from "1.21 V" to "1.25 V".
00003850A (02-16-21)		initial release

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